

The Difference a Day Makes: Timely Disclosure and Trading Efficiency in the Muni Market*

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Abstract

We study the impact of more timely disclosure on trading efficiency in over-the-counter markets. On January 31, 2005, the *Real-Time Transaction Reporting System* (RTRS) reduced the delay in reporting municipal bonds trades from one-day to 15 minutes. We provide evidence that the RTRS resulted in a significant reduction in overall trading costs for both large and small trades. Retail investors benefited primarily from the reduction in dealer intermediation costs, while sophisticated institutions realized large reductions in estimated bargaining costs. Most traded bonds experienced significant increases in trading volume. Consistent with greater market-making incentives by dealers, we find significant increases in dealer capital commitment, the length of intermediation chains, and a large reduction in pre-arranged trades. Overall, our findings support search- and inventory-based models' predictions regarding the impact of improved transparency on trading efficiency.

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1. Introduction

Many securities trade on decentralized, dealer-intermediated, over-the-counter (OTC) markets. Information regarding orders and bid/ask quotes, commonly referred to as pre-trade transparency, is often unavailable in these markets. Post-trade transparency is also commonly limited due to the slow public dissemination of transaction prices and volumes. The lack of price and trade transparency is frequently cited as an important reason why OTC markets suffer from low liquidity and high trading costs. In this paper we study the impact of improvements to post-trade transparency in the municipal secondary market. The \$4 trillion OTC municipal bond market has traditionally been characterized as opaque, fragmented, illiquid and one that imposes high trading costs on its diverse set of retail and institutional investors.¹ As a result, our evidence has policy relevance for retail and institutional investors, municipalities concerned about their borrowing costs, and regulatory rule making.

On January 31, 2005, with the explicitly stated purpose of increasing price transparency and enhancing surveillance and auditing by enforcement agencies, the Municipal Securities Rulemaking Board (MSRB) implemented the *Real-time Transaction Reporting System* (RTRS). The RTRS reduced the delay in trade reporting, for virtually all municipal bond trades, from one day to within 15 minutes of trade execution.² We find a significant reduction in transaction costs in the municipal secondary market after the introduction of RTRS. Furthermore, we find evidence to support both search- and inventory-based explanations for the reduced costs that appear to result from timelier trade reporting.

Our results contrast and complement Schultz (2012) that finds no decline in transaction

¹ For example, Harris and Piwowar (2006) show that poor information quality is the primary reason that municipal bond trades are significantly more expensive than equivalent-sized equity trades.

² See Rule G-14 on www.msrb.org and SEC file on www.sec.gov. SEC order: Release No. 34-50294.

costs in the municipal primary market after the introduction of RTRS. We believe that a crucial factor in reconciling these distinct findings is that the primary market is more likely to be characterized as a low search cost environment relative to the secondary market where few trades and stale information prevail. Therefore, the RTRS is expected to have a larger impact on the informational environment and trading efficiency in the secondary market than in the primary market. Our paper also complements studies on changes in corporate bond transparency (e.g. Schultz (2001); Bessembinder, Maxwell, and Venkataraman (2006); Goldstein, Hotchkiss, and Sirri (2007); Bessembinder and Maxwell (2008)). These studies focus on trading costs in the corporate bond market as trade reporting became more transparent through the Trade Reporting and Compliance Engine (TRACE) system.³ Furthermore, Schultz and Song (2019) find trading costs declined in the mortgage backed securities market after more timely post-trade disclosures were introduced in 2012, and this impact was larger among less central dealers. By expanding the scope of research to include changes in various components of trading costs, trading activities, and dealer inventory, our paper is able to shed light on the channels through which transparency impacts trading efficiency.

We begin by investigating the impact of RTRS on overall trading costs and intra-day price dispersion and whether cost savings vary by investor sophistication. Using comprehensive secondary market transaction records from the MSRB, we follow the methodology in Green, Hollifield, and Schürhoff (2007b) to estimate the total round-trip transaction costs, or dealer markup, by matching each dealer purchase with subsequent dealer sales for March-December 2004 (pre-RTRS period) and March-December 2005 (post-RTRS period). We find that dealer markups declined after the implementation of RTRS by an average of 30 basis points or 14%. Using trade

³ Like TRACE, the muni markets' RTRS made timely knowledge of trading prices easier to obtain. Thus market makers and dealers no longer possessed private price information (beyond 15 minutes) of a trade.

size as the proxy for investor sophistication, we find significant declines across all trade size groups. Markups for small and medium sized trades declined by 20 to 33 basis points, compared to a reduction of about 9 basis points for the largest trades. However, the decline in trading costs is largest in percentage terms for the largest trades. Like Shultz (2012), we observe a significant reduction in intra-day price dispersion across all trade size groups. Overall, the evidence suggests that both retail and institutional investors benefited from the improvement in post-trade transparency.

A natural concern in interpreting our results is that OTC markets are likely to have changed in other ways during the 2004-2005 time period. For example, the increasing use of electronic communications, internet usage, and alternative trading platforms could have increased informational efficiency in all OTC markets. To control for common improvements to the OTC markets, we build a control group from a subset of corporate bonds that began disseminating their transaction information in real time through the first two phases of the TRACE program. We choose bonds in the first two phases of TRACE to ensure that our control group bonds had time to adjust to changes in transparency brought about by TRACE.⁴ After controlling for markups in the corporate bond market, we continue to find that the RTRS significantly reduced trading costs in the muni market across all trade size groups. To the extent that liquidity spillovers occurred from the corporate to the muni market⁵, we would expect our estimates to understate the benefits of greater post-trade transparency in the bond markets.

Next we explore the theoretical mechanisms through which timelier reporting and trading

⁴ The TRACE program was implemented in three phases between July 2002 and February 2005. See Section 5.2 for a detailed discussion on the time line and the types of corporate bonds disseminated through each phase.

⁵ Bessembinder, Maxwell and Venkataraman (2006) develop a model and find evidence that when bond transparency improves for a subset of bonds, greater price precision among the subset better predicts the prices of bonds that do not directly benefit from price transparency. These “non-transparent” bonds indirectly benefit because their prices can be more accurately predicted with the subset of better prices. This phenomenon is referred to as a liquidity spillover.

costs are likely to be related. The theoretical literature identifies search frictions and inventory risk as two important factors that affect trading efficiency in OTC markets. Assuming symmetric information and no inventory risk, Duffie, Gârleanu, and Pedersen (2005, 2007) and Lagos and Rocheteau (2009) show that a reduction in search frictions increases investors' outside options, because it is easier to locate alternative trading counterparties, and thus improves their bargaining positions relative to dealers.⁶ With heterogeneous investors, Duffie, Gârleanu, and Pedersen (2005) further show that sophisticated investors benefit more from the reduction in search frictions due to their better access to dealers. In our setting, better post-trade disclosure brought about by the RTRS could attract more investors to the muni market and thus make it easier for dealers to match orders. It also provides investors a timely reference on transaction terms and thus a better bargaining position during negotiation with dealers. The empirical prediction is that better post-trade transparency can lower search costs, reduce the ability for dealers to extract rents from investors, and thus lead to lower bid/ask spreads (markups).

The predicted impact of lower search costs on trading activities is less clear. To the extent that lower search costs attract more investors to the market, better post-trade transparency can speed up trades and therefore increase trading volume. However, Duffie, Gârleanu, and Pedersen (2005) caution that shrinking profit margins for dealers, which result from the weakening of their bargaining power, can reduce their incentives to commit capital and participate in market making, especially in less liquid security markets.⁷ The reduced market-making by dealers can result in a decrease in the overall trading activities in the muni market.

⁶ Green, Hollifield, and Schürhoff (2007b) show that dealers' bargaining power represents a large part of the markups they charge to investors. Li and Schürhoff (2019) construct a dealer network and show that it has a core-periphery structure where dealers near the center of the network are more efficient at executing trades but are also more expensive than peripheral dealers. We focus on the impact of greater transparency on the relative bargaining power between dealers and investors and show that the effect varies by trade size which we interpret as a proxy for investor sophistication.

⁷ Bloomfield and O'Hara (1999) provide experimental evidence that trade disclosure increases informational efficiency of prices but increases spreads.

Inventory-based models focus on the impact of asymmetric information and inventory risk on trading costs. Naik, Neuberger, and Viswanathan (1999) show that better post-trade disclosure reduces adverse selection and improves dealer incentives to share inventory risk. As a result, investor welfare benefits from lower transaction costs in liquidity-driven trades. For information-driven trades, greater transparency fully reveals the private information learned by one dealer from trading with the investor. Similar to Hirshleifer (1971), in equilibrium the informed investor is forced to bear more of the price revision risk since the initial dealer loses the information advantage over other dealers. Hence, the impact of better post-trade disclosure on investor welfare could depend on trading motives. Improved risk sharing can also increase dealers' inventory capacity and thus lead to an increase in the overall trading activities in the muni market.

Motivated by these theoretical predictions, we explore tests of the search- and inventory-based explanations. First, we decompose dealer markups into *intermediation costs* and *bargaining power gains* by estimating a structural model developed by Green, Hollifield, and Schürhoff (2007b). The intermediation cost component compensates dealers for inventory risk taking. The bargaining power component captures dealers' ability to extract trading surplus from customers. We examine these two cost components across different trade sizes under the assumption that trade size is a proxy for trading motives (liquidity-driven vs. information-driven) and investor sophistication (i.e., larger sized trades are more likely information driven and carried out by sophisticated investors).

We find significant variation in the effects of RTRS on intermediation and bargaining costs across trade size. For smaller trades (less than \$200,000), intermediation costs dropped about 10 to 17 basis points, compared to a smaller increase for the larger trade size groups. Under the assumption that smaller trades are more likely to be liquidity driven, this evidence is consistent

with the inventory-based model in Naik, Neuberger, and Viswanathan (1999). Interestingly, changes in bargaining cost estimates vary by trade size with a U-shape pattern. Medium sized trades (between \$200,000 and \$1,000,000) display the largest reductions in bargaining costs, dropping by an estimated 24 to 25 basis points. In contrast, our estimates of reduced bargaining costs for smaller trades (less than \$200,000) and the largest trades (above \$1,000,000) are smaller. Consistent with search-based models in Duffie, Gârleanu, and Pedersen (2005, 2007) and Lagos and Rocheteau (2009), better post-trade disclosure improves the overall bargaining position of investors but the effects differ by investor sophistication. Unsophisticated retail investors have little improvement in their bargaining positions. The medium sized trades are likely carried out by wealthy individuals or average size institutions. These investors are likely sensitive to and involved in active negotiation with dealers. The availability of timely trading information after the RTRS improved their bargaining positions and led to a reduction in dealers' ability to extract rents. The largest, and presumably most sophisticated traders, were better informed and had strong bargaining positions even before the implementation of RTRS. Therefore, it is not surprising to observe only a modest improvement in transaction terms following the RTRS.⁸

In an effort to better understand the channels through which the RTRS impacted trading efficiency, we next look beyond markups and examine changes in other dimensions of trading including trading volume, dealer capital commitment, and number of trades in the intermediation chain. If greater transparency reduces search frictions and improves risk sharing, both the inventory- and search-based models predict more active trading, higher capital commitment for inventory imbalances, and more interdealer trades to complete a round trip. However, critics argue

⁸ Unlike many findings in the municipal bond trading literature, these appear to be consistent with the equity market. In particular, Barclay and Warner (1993) find that medium sized equity trades are where informed traders are able to best execute their informed "stealth" trades.

that mandatory transparency is not likely to add much value to actively traded securities but could have negative impact on thinly traded ones.⁹ If dealers face shrinking margins, the RTRS may reduce dealers' incentives to commit capital to market-making activities – especially for less liquid securities.¹⁰

Across the liquidity spectrum, we find no empirical evidence that these theoretically motivated drawbacks to transparency impact the trading activities following the implementation of RTRS. Moreover, following Bessembinder, Jacobsen, Maxwell, and Venkataraman (2018), we estimate the overnight and over-the-week capital committed to inventory for each bond across all dealers. In the post-RTRS period, we find that dealer capital commitment increased for most bond liquidity groups. In addition, more dealers were involved in completing a round trip transaction, resulting in an increase in the length of intermediation chains. Finally, dealers arranged more roundtrip transactions that involve taking on inventories while reducing the proportion of pre-arranged trades¹¹ across all trade size groups. Because pre-arranged trades involve virtually no inventory risk, our finding supports the interpretation that improved transparency leads to better inventory risk sharing among dealers. These results collectively suggest that market liquidity improved following the RTRS as dealers committed more capital to market-making activities.

The remainder of the paper is organized as follows. Section 2 provides a brief overview of the information environment in the municipal bond market. Section 3 presents the data and summary statistics. In Section 4, we briefly describe the methodology used to estimate and decompose trading costs. In Sections 5 and 6, we assess the impact of RTRS on dealer markups

⁹ For example, Asquith, Covert, and Pathak (2013) show that high yield corporate bonds publicly disseminated through TRACE in the final phase experienced a 41.3% reduction in trading volume.

¹⁰ Schultz and Song (2019) find evidence of lower capital commitment and less interdealer trading after more timely trade disclosures were introduced in the mortgage backed securities market.

¹¹ Pre-arranged trades are trades where the buy and sell occur at precisely the same time in the EMMA transaction database and reflect a negotiated off-market trade between two parties in which the dealer does not have to take an inventory position.

and investigate the search- and inventory-based explanations. Section 7 concludes.

2. Transparency and Disclosure Requirements in the Municipal Bond Market

The municipal bond market suffers from relatively small issue sizes combined with heterogeneity in virtually every important dimension. Potential municipal issuers include approximately 44,000 state and local governments, school districts, redevelopment agencies, and public utility districts. Municipal credit analysts must consider differing sources of repayment¹², credit ratings, redemption provisions, credit enhancement, tax-exemption status at the state and Federal level, among other factors in valuing municipal debt.¹³ These differences complicate bond valuation and result in a diverse and fragmented set of potential investors. At year-end 2011, individual investors directly held about half of the outstanding municipal bonds and mutual funds held about 25% on behalf of individual investors. Insurance companies and commercial banks collectively held about 20% of outstanding municipal bonds.¹⁴

Although the municipal securities market is often described as a “buy-and-hold” market, trading in the secondary market is significant. As shown in Table 1, the average daily trading volume was about \$8 billion in 1999, reached a peak over \$25 billion in 2007, and declined after the financial crisis to less than \$9 billion in 2015. For reference, municipal trading volume was actually higher than the corporate bond market between 2005 and 2008 but declined steadily after the financial crisis to about 30% of corporate bond volume in 2015. Dealers play a pivotal role in

¹² Depending on the source of financing, municipal bond issues can be generally classified as either tax supported general obligation bonds or revenue supported project bonds. General obligation bonds are backed by the general credit and taxing power of the state or local government. Revenue bonds are generally issued to finance specific revenue-generating projects such as toll-generating roads, bridges, airports, water and sewage facilities, health care and housing projects.

¹³ The majority of municipal bonds have the advantage of providing investors with coupon income that is exempt from Federal tax and often state taxes as well. Taxable municipal bonds and bonds subject to the alternative minimum tax add additional complication to the evaluation of bond value.

¹⁴ “Report on the Municipal Securities Market,” SEC, July 2012, page 14.

municipal bond trading. There are more than 1,800 municipal bond dealers but dealer market share is highly concentrated, with the top 10 dealers accounting for about 75% of the total par value traded.¹⁵ Traditionally a lack of transparency has challenged this dealer-intermediated market, especially for retail investors who have more limited access to information. In 2011, retail size trades (up to \$25,000) accounted for about 1/3 of the total transactions or about \$58 billion in par value.¹⁶

Since the mid-1990s, the MSRB has adopted a series of reporting requirement changes to improve the transparency in the municipal bond market. Appendix A provides a detailed timeline of these changes. In the early stages, trade dissemination requirements were limited to frequently traded bonds with a one-day lag. The MSRB started to release transaction information for infrequently traded bonds in October 2000, but with a one-month lag. By the end of 2002, the MSRB shortened the delay to one week and relaxed the requirement for frequently traded bonds to two or more transactions per day. On June 23, 2003, the MSRB introduced the enhanced “T+1 Daily Reports” disseminating transaction details for all bonds with a one-day lag.¹⁷ On January 31, 2005, the MSRB implemented the *Real-Time Transaction Reporting System* (RTRS), which was approved by the SEC on December 8, 2004. The RTRS requires dealers to report all transactions within 15 minutes of trade execution and the MSRB immediately releases the transaction report to the public.

¹⁵ “Report on the Municipal Securities Market,” SEC, July 2012, page 20.

¹⁶ “Report on the Municipal Securities Market,” SEC, July 2012, page 2.

¹⁷ The enhanced T+1 Daily Report will display exact par value for all trades with a par value of \$1 million and under. It displays “\$1 MM+” for trades that have a par value greater than \$1 million.

3. Data and Summary Statistics

The MSRB's intraday transaction database reports the bond identifier (CUSIP), transaction date and time, par value traded, and transaction price for all municipal bond trades. The database also identifies whether each transaction is a dealer buy, a dealer sell, or an interdealer trade. We use the Mergent Municipal Bond Database to obtain key bond characteristics such as bond type, offering date, offering size, maturity date, coupon rate, issuing state, credit rating, redemption provisions, etc. We merge these two databases by CUSIP.

To focus on the most relevant transaction data, we eliminate all puttable bonds and variable rate bonds. Variable rate bonds account for more than 40% of the dollar volume in the transaction database due to remarketing on interest rate reset dates (Harris and Piwowar 2006). These remarketings are periodic (usually weekly) on interest rate reset dates and trades almost always occur at par value, which is not surprising given that the remarketing agent is charged to reset the new weekly interest rate in order to sell the bonds at par. We exclude these trades since the zero-price markups do not reflect transaction costs. We also exclude all pre-arranged trades from the change analyses of trading volume, dealer capital commitment, and length of intermediation chains. These trades are pre-negotiated by dealers matching the buyer with the seller and thus bear virtually zero inventory risk. We identify these trades by matching buy and sell trades with the same CUSIP, par value, and time stamp. We discuss the details of matching procedure for identifying pre-arranged trades in Section 6.3. In addition, we eliminate transactions from our analysis with missing trade types, obvious errors, and large temporary price deviations. For example, we delete records with negative prices and transactions with par values less than \$5,000. For interdealer trades, transactions are in many cases reported twice by both the seller and the buyer. We sort all trades by their time stamps and keep the average price for consecutive

interdealer trades matched by the CUSIP and par value. Since our focus is on transactions in the secondary market, we follow Green, Hollifield, and Schürhoff (2007b) and delete transactions that occur in the first 90 days after issuance.

Table 2 provides summary statistics for daily transactions in the secondary municipal bond market during each two-year period from January 1999 to December 2012. As shown in Panel A, the average number of daily transactions increases over time from less than 10,000 in the early part of our sample period to over 20,000 by the end of 2012. The average number of distinct municipal bonds traded increases over the sample period but accounts for less than 1% of outstanding bonds. The relatively sparse trading of muni bonds is well documented in prior studies of the municipal market. Panel B shows that, on a typical trading day, customer purchases make up the largest proportion of trades (about 40-56%), followed by customer sales (about 23-29%) and inter-dealer trades (about 18-33%).

We categorize trade sizes at or below \$200,000 as small, between \$200,000 and up to \$1,000,000 as medium, and above \$1,000,000 as large. We further refine each category into smaller groups to capture the potential within-category heterogeneity: \$0-\$25,000, \$25,000-\$100,000, \$100,000-\$200,000, \$200,000-\$500,000, \$500,000-\$1,000,000, and above \$1,000,000. Consistent with Schultz (2012), in Panel C of Table 2, we find that retail size trades (less than \$25,000) account for the largest proportion of trades over our sample period – more than half of the total number of daily trades. In contrast, Panel D shows that the largest trade size group (above \$1,000,000) comprises 45-60% of the total dollar volume of daily trading but accounts for less than 3% of the total number of daily trades.

4. Estimation and Decomposition of Dealer Markups: Methodology

Following Green, Hollifield, and Schürhoff (2007b), we estimate dealer markups and their components. A brief description follows, and we refer readers to the original source for more complete details.¹⁸

4.1. Estimating Dealer Markups

Using all unique bonds sorted by trade time, we estimate dealer markups by pairing bond purchases and sales using the following algorithm. First, we check whether the initial dealer purchase from a customer is followed by a dealer sale to a customer for the same bond, for exactly the same par amount, on the same day, with no intervening purchases from customers.¹⁹ We call this an “*Immediate Match*.”²⁰ Next, we examine cases in which the initial dealer purchase is split into several smaller sales to customers, again with no intervening purchases from customers. We aggregate the subsequent consecutive sales until the aggregated par amount is equal to the initial purchase. We refer to this sequence of trades as a “*Roundtrip Match*.” For dealer purchases not categorized as *Immediate Match* or *Roundtrip Match*, we apply the first-in-first-out (FIFO) rule to deal with intervening purchases from customers. Specifically, we assign the first sales to the bonds that came into dealer inventories first. We refer to this sequence as a “*FIFO Match*.”²¹

Using these matched trading pairs, we calculate the gross dealer markup for each pair as the ratio of the difference between the par-weighted average sale price and the dealer’s purchase

¹⁸ See in particular, Sections 2 and 3 of Green, Hollifield, and Schürhoff (2007b).

¹⁹ One challenge in estimating dealer markups is that the MSRB transaction data does not provide unique identification for broker-dealers and customers. However, given that municipal bonds trade so infrequently, it is unlikely that different dealers are trading the same bond at the same time with their customers.

²⁰ Within the immediate match group, in studying inventory costs (Section 6.3) we will exploit pre-arranged trades where time stamps are identical as well.

²¹ We exclude FIFO matches that are not completed within a 30-day window from the initial trade to focus on bonds held in inventory for limited periods. Although the FIFO method could mistakenly assign sales by one dealer to purchases made by another dealer, one can interpret the estimated markups as the representative profits to the dealers as a group.

price to the purchase price, or

$$\text{Gross Markup} = \frac{\text{Par-weighted average sale price} - \text{Purchase price}}{\text{Purchase price}}. \quad (1)$$

Since the matched dealer sales may be days away from the initial purchase day, we adjust the gross markup for market movements. We capture market movements by using daily returns for the Barclays' Municipal Bond Index for 1, 3, 5, 7, 10, 15, 20, and 22+ year maturities from Morningstar Direct. We subtract the maturity-matched return of the corresponding municipal bond index from the gross markup to obtain the net dealer markup:

$$\text{Net Markup} = \text{Gross Markup} - \text{Return on Maturity-matched Index}. \quad (2)$$

Net Markup is our estimate of the investor's round-trip transaction cost, since it captures the bid-ask spread for the dealer.

4.2. Decomposing Dealer Markups

The markup estimated in (2) reflects the costs to dealers for intermediating trades (i.e., inventory risk) and the ability of dealers to extract trading surplus relative to customers (i.e., bargaining power). We use a structural model to decompose the dealer markup into the *Intermediation Costs* and *Bargaining Power Gain* components as follows:

$$\text{Net Markup} = \text{Intermediation Costs} + \text{Bargaining Power Gain} + \text{Unexpected Error}. \quad (3)$$

In Appendix B, we present the details of our estimation model (3) that includes only minor modifications of the Green, Hollifield, and Schürhoff (2007b) model. To parameterize *Intermediation Costs*, we use the linear specification:

$$\text{Intermediation Costs} = \theta_0 + \theta_{01} \text{RealTime} + \theta \mathbf{X}, \quad (4)$$

where $\mathbf{X}$ is a vector of bond and trade characteristics, θ is the corresponding vector of coefficients, and θ_0 is the constant term. To capture the effect of RTRS on the intermediation costs, we add an

indicator variable *RealTime* to the constant term that equals one if the trade occurs post-RTRS and zero otherwise. The coefficient θ_{01} therefore reflects the change in average intermediation costs after the implementation of RTRS.

The *Bargaining Power Gain* component is modeled with a *one-sided* error term to capture the dealer's incentive compatibility constraint for each trade. Specifically, we assume that the dealer's gain from bargaining power, conditional on the realization of positive total trading surplus, follows an exponential distribution. We parameterize the conditional mean of dealer's trading surplus as a log-linear function of bond and trade characteristics:

$$E(\text{Bargaining Power Gain} | \text{Positive Trading Surplus}) = (a_0 + a_{01} \text{RealTime}) * e^{aZ}, \quad (5)$$

where Z is the vector of explanatory variables, a is the corresponding vector of coefficients, and a_0 is the constant term. We add the *RealTime* indicator variable to the constant term and its coefficient a_{01} should capture the impact of RTRS on the dealer's share of trading surplus in a typical transaction.

We simultaneously estimate the coefficients in structural model (3) using maximum likelihood. The definition of all explanatory variables included in the model can be found in Appendix C. To facilitate the economic interpretation of coefficients, we subtract the sample means from all explanatory variables. The intercept terms therefore represent the average dealer markup in each component for a typical trade in basis points. We are particularly interested in the coefficients for the *RealTime* indicators that capture the impacts of RTRS on the intermediation costs and bargaining power gain components, estimated respectively from (4) and (5).

5. The Impact of RTRS on Dealer Markups

We conduct a series of empirical tests to assess the impact of RTRS on dealer markups. First, we present the mean test results on the change of dealer markups before vs. after the implementation of RTRS. We then estimate a pooled regression controlling for various bond, issuer, and trade characteristics. Finally, we conduct a diff-in-diff analysis using a subset of corporate bonds as the control group to help evaluate causality.

5.1. Mean Tests on the Change of Dealer Markups

Using the algorithms outlined in Section 4, we estimate the dealer markups for all round-trip transactions in the MSRB database from 2002 to 2012. We study a number of windows surrounding the January 31, 2005 implementation date and in all cases we drop transactions that occurred in January and February of 2005 to mitigate the effects of transitioning to the new reporting system. We delete outliers by truncating the estimated dealer markups at the 0.5th and 99.5th percentiles. We focus our discussion on the ten-months preceding and the ten-months succeeding the implementation of RTRS.²² The final sample includes 449,534 dealer purchases for the 226 trading days from March 2004 to December 2004, and 443,907 dealer purchases for the 214 trading days from March 2005 to December 2005.²³ In Panel A of Table 3, we report average dealer markups in each time period for three samples. The “All Bonds” sample includes all bond transactions where we estimate trading costs. The “Same Issuers” subsample includes only bonds issued by the same set of issuers in both time periods. The “Same Bonds” subsample

²² Our results are robust to the time window chosen. A prior version of our paper focused on approximately two-year windows, and we have also estimated changes in the prior and succeeding six-month windows. All of the windows we have examined lead to the same qualitative conclusions.

²³ Trading days are different because some bonds are reported to trade on weekends and we include those days as trading days if there are sufficient trades.

includes only bonds that are traded in both time periods.

Of primary interest, we observe a significant decline in average dealer markups after the implementation of RTRS. The mean markup for the “All Bonds” sample declined from about 209 basis points in the pre-RTRS period to 179 basis points in the post-RTRS period, a reduction of 30 basis points or 14%. Moreover, we find large declines in average trading costs across all trade size groups that are all statistically significant at the 1 percent level. In general, the smaller sized trades showed larger absolute reductions in trading costs but smaller percentage changes than the larger trades. The average trading costs for the smallest trades ($\leq \$25,000$) fell from 244 to 215 basis points, a 12% drop. In contrast, the average trading cost for the largest trades ($> \$1,000,000$) fell by 9 basis points or a 30% drop. Figure 1 plots the average monthly dealer markups for each trade size group before vs. after the implementation of RTRS. The results are similar if we use the median, par-value weighted, and market-value weighted markup measures. Consistent with prior empirical findings²⁴, average dealer markups decrease monotonically with trade size in both time periods. Larger sized trades are likely carried out by sophisticated institutional investors, who incur lower trading costs due to better information and more bargaining power.

Panel B of Table 3 examines the change in price dispersion before vs. after the implementation of RTRS. Following Schultz (2012), we compute the price dispersion as the standard deviation of intra-day trading prices for each bond with a minimum of two trades on any given trading day. We then average across all bonds traded on that day to obtain the daily price dispersion measure. We next take the average across all trading days in the pre- and post-RTRS periods and test the difference in means. A high daily price dispersion indicates that investors pay a wide range of prices for the same bond trading on the same day – signaling a high degree of

²⁴ See Downing and Zhang (2004), Harris and Piwowar (2006), Edwards, Harris, and Piwowar (2007), and Green, Hollifield, and Schürhoff (2007a, 2007b).

market segmentation. More timely trade disclosure is expected to reduce the daily price dispersion as investors become more informed about the terms of transactions occurring earlier in the day. We find that the average daily price dispersion for the “All Bonds” sample declines from \$0.76 in the pre-RTRS period to \$0.65 in the post-RTRS period, a reduction of \$0.11 or about 14%, which is statistically significant at the 1% level. Similar to the changes in transaction costs, we observe a significant reduction in price dispersion across all trade size groups. The smaller trades show larger reductions in absolute values but the larger trades benefit from a larger percentage decrease. Consistent with Schultz (2012) and Green, Hollifield, and Schürhoff (2007a), we observe much smaller price dispersion for larger trades than for smaller trades. This suggests that retail investors are less informed and may benefit more from better disclosure.

The results on the changes of dealer markups and price dispersion are similar for the “Same Issuers” and “Same Bonds” subsamples. This suggests that these changes in trading costs and price dispersion are not due to changes in the composition of the types of bonds that are being traded.

5.2 Pooled Regression and Diff-in-Diff Analysis

The main empirical challenge we face is that the RTRS is a policy change with a single-event date. We undertake two additional analyses in an effort to better identify the impact of RTRS on trading costs.

In Table 4, we estimate a pooled regression of dealer markups on a battery of control variables. We include all secondary market transactions in the two ten-month test periods in 2004 and 2005. As control variables, we include trade, bond, and issuer characteristics such as number of interdealers involved, par value traded, maturity, coupon, age, issue size, bond type, credit rating, redemption provision, insurance provision, issuer type, etc. In addition, we control for overall

market conditions measured by treasury yield and treasury yield volatility. We also include issuer fixed effects and cluster the standard errors by bond. The variable of interest is the *RealTime* indicator that equals one if the transaction occurred after the implementation of RTRS and zero otherwise. The coefficients on *RealTime* are negative and statistically significant across all trade size groups. Moreover, the magnitude of changes in dealer markups decline in trade size, ranging from about 17 basis points drop for the smallest sized trades ($\leq \$25,000$) to 4 basis points for the largest sized trades ($> \$1,000,000$). These results are consistent with the mean tests reported in Table 3. The coefficients for control variables are largely consistent with the results in Green, Hollifield, and Schürhoff (2007b). Regarding the impact of intervening interdealer trades on trading costs, we find that more interdealer trades increase markups (similar to Schultz (2012)) but significantly less so after the introduction of RTRS. The declines are economically large (10 to 20 basis points) and statistically significant at the 1 percent level across all trade size groups.

In a second test to isolate the impact of RTRS from unobserved common bond market transaction cost factors, we use a subset of corporate bonds as a control group and apply the same markup algorithm to estimate corporate bond markups using TRACE data. The implementation of TRACE improved trading transparency in the corporate bond market from 2002 to 2005.²⁵ With TRACE, the dissemination of post-trade transaction information took place in three phases. Phase I was implemented on July 1, 2002 and included investment-grade bonds with an initial issue size of \$1 billion or greater.²⁶ Phase II was implemented on March 3, 2003 and applied to investment-grade bonds of at least \$100 million initial issue size or bonds with ratings of A- or higher. On April 14, 2003, TRACE expanded to include another 120 investment-grade bonds rated BBB. All

²⁵Edwards, Harris and Piwowar (2007) and Bessembinder, Maxwell, and Venkataraman (2006) each provide a more complete description of the rollout of TRACE.

²⁶Phase I also included 50 high yield bonds under the Fixed Income Pricing System (FIPS). We exclude these bonds from the control group.

remaining corporate bonds were included in TRACE in Phase III between April 22, 2004 and February 7, 2005.

Mindful of the rollout of TRACE, and in order to best isolate the common bond market factors that could drive down transaction costs, we include in our control group only those corporate bonds that became subject to TRACE reporting in the first two phases (about 4,600 bonds). This provides comfort that the corporate bonds we use as controls were not simultaneously undergoing a change in transparency. Since Phase II of TRACE included its last cohort of bonds in April 2003, this allows about 11 months prior to the start of our pre-RTRS period in March of 2004, for the control sample of corporate bonds to reflect the benefits of improved trading transparency. For compatibility, we also restrict the municipal bonds in the treatment group to have the same credit ratings and minimum issue size as the Phases I and II corporate bonds. Figure 2 plots the monthly average dealer markups for the treatment group and the control group in the pre- and post-RTRS periods. The figure suggests similar declines in muni and corporate bond dealer markups in the pre-RTRS period, while post-RTRS, it appears that average markups in the municipal market continue to decline while corporate markups level out.²⁷ To formally test these observations, we use corporate bond transaction costs as controls and conduct a diff-in-diff regression with the following specification:

$$Net Markup_{i,t} = Constant + \beta_1 Muni_i + \beta_2 RealTime_t + \beta_3 Muni_i * RealTime_t + B_4 Control Variables_{i,t-1} + e_{i,t}. \quad (6)$$

Here we use i to denote each bond transaction and t to denote the month in which the transaction was completed. The dependent variable is the net markup for each round-trip transaction as defined in (3). $Muni$ is an indicator variable that equals one if the bond traded is a municipal bond and zero

²⁷ The correlation coefficient of average markups between the two markets is 0.94 (statistically significant at the 1% level) in the pre-RTRS period, compared to 0.31 (statistically insignificant) in the post-RTRS period.

for a corporate bond. *RealTime* is an indicator variable that equals one if the transaction occurred in the post-RTRS period and zero otherwise. The variable of interest is the interaction term *Muni*RealTime*, which captures the diff-in-diff effect of RTRS after controlling for the change in markups in the corporate bond market. We also include in the regression a set of control variables that are available and relevant for both municipal and corporate bonds. We obtain key corporate bond characteristics from the Mergent database. We estimate equation (6) for each trade size group and report the regression results in Table 5. Table 5 shows that the coefficient estimates for *Muni*RealTime* are negative and statistically significant for all trade size groups, suggesting that the decline in dealer markups post-RTRS in the municipal bond market was not caused by unobserved factors affecting the trading costs in both the corporate and municipal markets. We also find that an increasing number of interdealer trades raises markups more in the muni market than in the corporate bond market, but this effect declines after the introduction of RTRS.

In summary, we find that more timely disclosure of past transactions leads to economically significant reductions in secondary market trading costs. This complements and contrasts with Schultz (2012) that did not find evidence that dealer markups declined in the primary market. We believe that this difference can be reconciled by the fact that search costs are relatively low during a primary offering because information is timely and relatively plentiful, while in the secondary market search costs are much higher because financial information is stale and trading is sparse relative to the initial offering period.

6. Search- and Inventory-based Explanations for the Impact of RTRS

In this section we investigate possible channels through which the RTRS may impact trading efficiency. As discussed in the introduction, we focus on evaluating the role of search costs

and inventory risk sharing around the implementation of RTRS. In particular, we decompose dealer markups into different components reflecting search costs and inventory risk and separately investigate the impact of RTRS on each component. We also investigate the changes in trading activities, dealer capital commitment, the length of intermediation chains for a roundtrip transaction, and pre-arranged trades to assess the theory.

6.1 Impact of RTRS on the Components of Dealer Markups

Following the methodology described in Section 4.2 and Appendix B, Table 6 reports the maximum likelihood estimates for the coefficients in the *Intermediation Costs* (4) and *Bargaining Power Gain* (5) specifications. We are particularly interested in the coefficients for the *RealTime* indicators, which capture the differential impacts of RTRS on the two cost components. Similar to Green, Hollifield, and Schürhoff (2007b), we observe that the bargaining power component is a major part of the total transaction costs across all trade size groups. This highlights the power of dealers in an illiquid and segmented OTC market. The bargaining power components for small and medium sized trades (less than \$1,000,000) ranged from about 70 to 100 basis points during the pre-RTRS period. In contrast, the bargaining power component for the largest sized trades (above \$1,000,000) was just 40 basis points. This is consistent with the interpretation that sophisticated large institutional investors are better informed and are able to extract better transaction terms than smaller investors.

We find that the effects of RTRS on the two cost components vary greatly across trade size. Regarding intermediation costs, smaller sized trades (less than \$200,000) experienced a significant decline of 10 to 17 basis points. In contrast, the changes for medium and large sized trades (above \$200,000) were positive but smaller in magnitude. Since smaller sized trades are more likely to be

liquidity driven, this finding supports the prediction of Naik, Neuberger, and Viswanathan (1999) that greater transparency improves the sharing of inventory risk among dealers. The change in bargaining power component exhibits a very different pattern. Smaller sized trades (less than \$200,000) had a small drop of 4 to 7 basis points. In contrast, for the medium sized trades (between \$200,000 and \$1,000,000), the decline was about 24 to 25 basis points and statistically significant at the 1 percent level. For the largest trades, we find a modest drop of about 12 basis points. The overall decline in the bargaining power component is consistent with the prediction of search-based theoretical models (Duffie, Gârleanu, and Pedersen 2005, 2007; Lagos and Rocheteau 2009). To the extent that greater transparency reduces search frictions, investors have better outside options and thus improve their bargaining positions when negotiating with dealers.

The U-shape change in the bargaining power component is particularly interesting. It suggests that the RTRS has a differential impact on leveling the playing field between dealers and different investors. More post-trade transparency benefits unsophisticated retail investors mainly through reducing the intermediation costs but has only a small impact on reducing dealers' bargaining power. Retail investors could be ignorant to new information or face high switching costs even in the presence of better trading opportunities. The medium sized trades appear to benefit the most from more bargaining power. Investors in this group are likely wealthy individuals and institutions, who are sensitive to new information and actively involved in negotiating with dealers. The availability of high quality trading information post-RTRS improves their bargaining positions and leads to a large reduction in dealers' ability to extract rents. For the largest and, presumably, most sophisticated institutions, they were better informed and had strong bargaining positions even before the implementation of RTRS. It is therefore reasonable to observe only a modest improvement in transaction terms following the RTRS. It is worth pointing out that these

results are generally consistent with findings in the equity markets that most informed “stealth” trading is concentrated in the medium sized trades (Barclay and Warner 1993, Chakravarty 2001).

We include the same set of controls as in Table 4 for other factors that help to explain the cross-sectional variation in dealer markups. For brevity, we report the coefficients in an online Appendix. In general, the controls behave as expected in that higher risk leads to higher net markups, though some do not have an obvious prediction. The coefficients are largely in line with the findings in Green, Hollifield, and Schürhoff (2007b).

6.2. Impact of RTRS on Trading Activities and Dealer Capital Commitment

To this point we have focused on the change in trading costs. However, improved transparency could also alter investors’ trading activities and dealers’ inventory positions. We examine trading volume and dealer capital commitment in an attempt to better identify the underlying mechanisms that are influenced by better post-trade disclosures. We are also interested in whether greater transparency has differential impact on trading activities for bonds with different liquidity. Proponents of more timely disclosure argue that greater transparency should lead to more active trading due to better risk sharing among dealers and lower search costs. Critics caution that greater transparency can actually reduce dealers’ incentives to provide market making services due to shrinking profit margins. The negative impact could be particularly strong for less liquid securities as a result of higher inventory risks.

We use trading volume to sort bonds into liquidity groups. For each municipal bond, we first compute monthly trading volume as the total dollar par value traded each month normalized by the issue size. If the bond does not trade in a certain month, the trading volume is recorded as zero. To address the concern that our rankings will induce mean reversion, we rank and define

each bonds' trading volume group in 2003, which is prior to the pre-RTRS period. We then calculate the baseline trading volume and capital commitment variables for each liquidity group in the pre-RTRS period. This approach ensures that the ranking period precedes and does not overlap with the pre-RTRS time period. We then assess the changes in trading activities and dealer capital commitment between the pre- and post-RTRS periods. Group 0 contains all municipal bonds that did not trade in 2003 and comprises about 45% of our sample. Conditional on observing at least one trade in the MSRB transaction database, we sort the remaining municipal bonds into quintiles (Groups 1 to 5). Table 7 presents summary statistics for each trading volume group. Comparing the no-trade bonds (Group 0) to the quintiles, the no-trade group tends to be older bonds with higher coupons, smaller issue size, and less insurance protections. Intuitively, less active trading groups tend to have higher dealer markups and larger price dispersion.

Change in Trading Activities

To study the impact of RTRS on trading activities, we estimate a diff-in-diff regression similar to (6) with the bond-level monthly trading volume as the dependent variable. We use the same treatment group of municipal bonds and control group of corporate bonds as in (6). Similar to the procedure used for municipal bonds, we sort corporate bonds into 6 groups based on the average monthly trading volume in 2003: the no-trade group (Group 0) and the quintile groups conditional on positive trading volume (Groups 1 to 5). We then run the diff-in-diff specification for each trading volume group and report the estimates in Table 8. The coefficient estimates for the interaction term, *Muni*RealTime*, are positive for all groups and statistically significant at the 10 percent level or lower. The magnitudes of changes are economically large as well. Relative to corporate bonds in the pre-RTRS period, municipal bonds with similar liquidity experienced an increase in monthly trading volume of 0.24%-0.78% after the implementation of RTRS.

Overall, we find no evidence that greater transparency reduces trading for less liquid bonds. In fact, the trading of muni bonds became more active for all liquidity groups following the implementation of RTRS. This is consistent with both better disclosure attracting more investors to the market and dealers providing more market-making services due to better inventory risk sharing.

Change in Dealer Capital Commitment

We next investigate the changes in dealer capital commitment for bond groups sorted based on trading volume in 2003. Inventory-based models suggest that greater transparency can improve the risk-bearing capacity of dealers due to better risk sharing. The empirical prediction is that dealers should be willing to commit more capital to absorb customer imbalances after the implementation of RTRS. In the spirit of Bessembinder, Jacobsen, Maxwell, and Venkataraman (2018), we calculate the aggregate dealer capital commitment for each bond overnight and over each week. Specifically, we calculate the absolute value of the dealers' aggregate buy volume less the dealers' aggregate sell volume during the day (or week), scaled by the total daily (or weekly) trading volume for the bond. These ratios can be respectively interpreted as the percentage of daily (or weekly) trading volume that is carried into overnight (or over the week) holdings. If no trading occurs during a certain horizon, we set the corresponding capital commitment measure to zero.

To control for overall market condition, bond characteristics, and other unobserved factors that may affect dealers' inventory risk taking, we estimate the following diff-in-diff regression with the same sets of muni and corporate bonds as in (6) during the pre- and post-RTRS periods:

$$D_{it} = \alpha + \beta_1 Muni_i + \beta_2 RealTime_t + \beta_3 Muni_i * RealTime_t + \beta_4 MarketCondition_{t-1} + \beta_5 BondChar_{i,t-1} + \varepsilon_{it}. \quad (7)$$

The dependent variable D_{it} denotes the aggregate overnight or weekly dealer capital commitment

for bond i in month t . We average across all days or weeks in a month to obtain the monthly measures. *Muni* is an indicator variable that equals one if the bond is a municipal bond and zero otherwise. *RealTime* is an indicator variable that equals one for the post-RTRS period and zero otherwise. The coefficient β_3 captures the change in average dealer capital committed to a municipal bond (relative to similar corporate bonds) post-RTRS. To control for the effect of overall market conditions, the vector variable *MarketCondition* includes the change in 1-month LIBOR, the change in VIX, the logarithm of total par value traded in the muni or corporate market, the Barclays maturity-matched muni or corporate index return, the standard deviation of daily Barclays index returns, the Treasury Yield, and the volatility of daily Treasury yield, all measured at the monthly frequency and lagged by one month. Of particular importance, the changes in LIBOR and VIX control for the impact of changes in overall funding costs on inventory risk taking. In addition, we include a vector of bond characteristics (*BondChar*) similar to the controls in Table 4.

Table 9 reports the overnight and weekly capital commitment regression results for bond groups sorted based on the 2003 trading volume. The coefficients on *Muni* are mostly negative and statistically significant, suggesting that dealer capital commitment is in general higher for corporate bonds. For both overnight and weekly capital commitment, the coefficients of *Muni*RealTime* are all positive and statistically significant except for Group 0. This suggests that dealers became more willing to absorb inventory imbalance post RTRS. The increases in daily and weekly capital commitment for Groups 1 to 5 ranged from 0.39% to 1.94% and from 1.08% to 2.07%, respectively. For Group 0, both the daily and weekly capital commitment declined post RTRS, but the magnitude was much smaller. Note that bonds in Group 0 were never traded in 2003 and remained lightly traded in the pre- and post-RTRS periods. These bonds were likely to

have low intrinsic trading demand from investors. Nonetheless, Table 8 shows that the reduced capital commitment in Group 0 is not associated with a decline in trading volume after the implementation of the RTRS.

Among the control variables, we find that dealer capital commitment is significantly higher for bonds with larger issue size, longer time to maturity, and lower coupon rates. Consistent with dealers reducing their inventory when funding costs are high, we find that increases in 1-month LIBOR predict significantly lower capital commitment. More importantly, the coefficients of *Muni*RealTime* remain significant after controlling for the impact of funding costs, suggesting that the impact of RTRS on dealers' inventory risk taking is unlikely driven by the overall trend of funding costs during the event period.

Overall, our results on trading activities and dealer capital commitment are consistent with inventory-based explanations. Timelier reporting and greater transparency in the post-RTRS period led to better risk sharing among dealers and increased market-making activities for municipal bonds.

6.3 Changes in the Length of Intermediation Chains and Proportion of Pre-arranged Trades

In our last two analyses, we investigate changes in the length of the roundtrip intermediation chains and the proportion of pre-arranged trades following the implementation of RTRS. We construct two measures for the length of the intermediation chains: the number of days and the number of interdealers involved in a round trip transaction. Table 10 reports the changes for the complete sample and by trading volume groups. We find that the average length of the intermediation chains increased significantly in the post-RTRS period for the complete sample. The average number of days to complete a round trip transaction increased from 3.23 days to 3.27

days, while the average number of interdealers involved increased from 0.36 to 0.41. Across different trading volume groups, we find significant increase in the average length of the intermediation chains for almost all groups. These results are consistent with inventory-based theories suggesting that reduced asymmetric information through more timely disclosure leads to better risk sharing among dealers and therefore more interdealer trading is undertaken.

In Table 11, we investigate the changes in proportion of roundtrip transactions for each match group: *immediate* match, *roundtrip* match, and *FIFO* match. Within the immediate match group, we identify a subset of *pre-arranged* matched trades where the time stamp, par size, and CUSIP are all identical for a buy and a sell. Pre-arranged trades represent off exchange negotiated trades matching the buyer with the seller and thus reduce inventory costs to (near) zero. The remaining trades in the immediate match group are classified as *same-day* matches. We find a striking reduction in the pre-arranged trades across all trade sizes, from 7.17% in the pre-RTRS period to 3.23% in the post-RTRS period for all bonds in the sample. Across the match categories, most of the reduction in pre-arranged trades is picked up by the roundtrip match trade category, from 58.78% to 63.53% for the complete sample. This provides further evidence that inventory costs become lower in the post-RTRS information environment, thus motivating dealers to engage in more risk-sharing rather than pre-arranged trades.

The analyses of the components of dealer markup, trading activities, dealer capital commitment, length of intermediation, and pre-arranged trades provide evidence consistent with trading efficiencies being realized through both the search- and inventory-based channels.

7. Conclusions

The implementation of the RTRS on January 31, 2005 marks an important change in the evolving transparency of the municipal bond market. Our paper provides evidence that the improved timeliness of post-trade disclosure has resulted in reductions in the costs and price dispersion encountered when trading municipal bonds. We also investigate the channels through which more timely reporting impacts trading efficiency and find supporting evidence for both search-based (Duffie, Gârleanu, and Pedersen 2005, 2007; Lagos and Rocheteau 2009) and inventory-based explanations (Naik, Neuberger, and Viswanathan 1999).

We find interesting variation in the impact of RTRS on the components of total transaction costs. Smaller trades benefit mainly from the reduction in intermediation costs, which is consistent with lower inventory costs due to better risk sharing among dealers. Medium sized trades benefit more from the gain in bargaining positions. This is consistent with greater transparency helping to level the playing field between dealers and sophisticated investors, as predicted by the search-based theories.

Further analyses on trading volume and dealers' market-making activities provide additional insights into the impact of improved transparency. We find a significant increase in trading volume, dealer capital commitment, and the length of intermediation chains in the post-RTRS period across most bond groups sorted based on liquidity. We also find that dealers significantly reduced the proportion of pre-arranged trades that involve no inventory risk and engaged more in facilitating round-trip transactions. These results support the inventory-based explanations that greater transparency improves the risk sharing among dealers and thus incentivizes more market-making activities.

Finally, changes in regulatory reporting requirements, such as the RTRS, carry costs.

Evaluating the impact of RTRS is critical to understand whether these reporting costs are providing benefits to investors. The adoption of RTRS appears to significantly reduce the trading costs to investors in the municipal secondary market. Ultimately, these reductions in investors' trading costs should be reflected in reduced borrowing costs for municipalities, since investors' required rates of return will reflect the lower costs of transacting (e.g. Amihud and Mendelson 1986). From a policy perspective, the costs of disclosing timely trading data to the public would have to be large to overwhelm the apparent benefits to municipal bond investors and issuers.

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Table 1 Aggregate Statistics for the Municipal and Corporate Bond Markets

Year	New Issuance (\$Billion)		Trading Volume (\$Billion)		Par Value Outstanding (\$Billion)		Total US Bond Market (\$Billion)
	Muni	Corporate	Muni	Corporate	Muni	Corporate	
1999	224.37	602.15	8.29		1,457.09	3,161.97	15,328.04
2000	198.24	575.06	8.77		1,480.71	3,447.16	16,171.71
2001	286.24	770.61	8.80		1,603.40	3,862.05	17,590.64
2002	355.72	636.74	10.72	17.83	1,762.82	4,066.54	18,982.12
2003	380.14	774.28	12.60	18.03	1,900.38	4,357.37	20,463.26
2004	358.06	775.78	14.80	17.29	2,874.27	4,583.02	22,905.90
2005	407.07	750.07	16.88	16.56	3,097.67	4,651.66	24,680.69
2006	385.87	1058.35	23.15	16.89	3,283.77	4,892.21	27,144.14
2007	429.16	1141.02	25.15	16.39	3,547.21	5,325.18	29,441.07
2008	389.26	716.97	19.37	14.32	3,662.90	5,501.49	31,056.65
2009	409.54	945.37	12.47	19.94	3,844.74	6,087.62	32,126.33
2010	433.13	1057.86	13.32	20.45	3,953.96	6,721.99	33,892.69
2011	295.06	1025.13	11.29	20.65	3,913.77	6,861.27	34,435.31
2012	382.63	1371.23	11.26	22.62	3,911.26	7,234.05	35,359.15
2013	335.30	1379.94	11.18	24.70	3,843.26	7,657.76	36,392.45
2014	339.13	1438.38	9.87	26.69	3,796.62	8,026.91	37,480.06
2015	405.11	1493.60	8.63	27.94	3,810.66	8,257.80	38,474.81

This table reports the aggregate annual statistics for the municipal and corporate bond markets from 1999 to 2015. New Issuance is the total par value of new bonds each year. Trading Volume is the average daily total par value of bonds traded during the year. Par Value Outstanding is the total par value of bonds outstanding at the year end. Total US Bond Market is the total par value of all fixed income securities outstanding in the US market at the year end, including interest bearing marketable coupon public debt, asset-backed securities, money markets, mortgage related securities, corporate debt, and municipal securities. The data source for aggregate US bond market is from the Securities Industry and Financial Markets Association (www.sifma.org).

Table 2 Summary Statistics for Municipal Bond Transactions

Items	1999-2000	2001-2002	2003-2004	2005-2006	2007-2008	2009-2010	2011-2012
Panel A: Overall Trading							
No. of Trades	9,983	10,023	10,262	11,117	15,582	20,949	20,223
No. of Bonds	226	265	304	363	410	449	429
No. of Issuers	21	25	27	30	32	34	32
Par Value Traded (\$MM)	1,993	2,245	2,306	2,676	3,100	3,588	3,549
Panel B: Number of Transactions by Trader Type							
Customer Purchase	5,520 (55.29%)	5,617 (56.04%)	5,222 (50.89%)	4,854 (43.67%)	7,270 (46.66%)	9,777 (46.67%)	8,479 (41.93%)
Customer Sale	2,569 (25.74%)	2,583 (25.77%)	2,936 (28.61%)	3,194 (28.74%)	3,639 (23.36%)	4,846 (23.13%)	5,091 (25.18%)
Inter-Dealer Trades	1,894 (18.98%)	1,823 (18.19%)	2,103 (20.50%)	3,068 (27.60%)	4,672 (29.98%)	6,326 (30.20%)	6,652 (32.89%)
Subtotal	9,983	10,023	10,262	11,117	15,582	20,949	20,223
Panel C: Number of Transactions by Trade Size							
0-25K	5,550 (55.59%)	5,710 (56.97%)	5,848 (56.99%)	6,007 (54.04%)	8,000 (51.34%)	11,427 (54.55%)	11,115 (54.96%)
25-100K	2,979 (29.84%)	2,979 (29.73%)	3,101 (30.22%)	3,561 (32.03%)	5,264 (33.78%)	6,850 (32.70%)	6,555 (32.42%)
100-200K	494 (4.95%)	476 (4.75%)	503 (4.90%)	600 (5.40%)	890 (5.71%)	1,088 (5.19%)	1,080 (5.34%)
200-500K	528 (5.29%)	487 (4.86%)	463 (4.51%)	532 (4.79%)	822 (5.28%)	962 (4.59%)	900 (4.45%)
500-1,000K	197 (1.98%)	180 (1.80%)	161 (1.57%)	183 (1.64%)	268 (1.72%)	302 (1.44%)	279 (1.38%)
1,000K+	234 (2.34%)	189 (1.89%)	185 (1.81%)	233 (2.10%)	338 (2.17%)	321 (1.53%)	293 (1.45%)
Subtotal	9,983	10,023	10,262	11,117	15,582	20,949	20,223

Table 2 (Continued) Summary Statistics for Municipal Bond Transactions

Items	1999-2000	2001-2002	2003-2004	2005-2006	2007-2008	2009-2010	2011-2012
Panel D: Par Value of Transactions by Trade Size (\$Mill)							
0-25K	81 (4.94%)	85 (6.26%)	91 (6.04%)	95 (5.09%)	127 (4.71%)	181 (7.19%)	181 (7.40%)
25-100K	191 (11.61%)	186 (13.61%)	202 (13.46%)	219 (11.73%)	326 (12.15%)	416 (16.57%)	412 (16.85%)
100-200K	88 (5.32%)	82 (5.99%)	90 (6.02%)	102 (5.43%)	149 (5.54%)	180 (7.15%)	184 (7.53%)
200-500K	202 (12.28%)	181 (13.30%)	180 (12.00%)	195 (10.43%)	291 (10.85%)	340 (13.54%)	327 (13.35%)
500-1,000K	192 (11.67%)	170 (12.43%)	158 (10.53%)	167 (8.91%)	239 (8.90%)	263 (10.48%)	250 (10.24%)
1,000K+	894 (54.19%)	660 (48.41%)	781 (51.96%)	1,093 (58.41%)	1,554 (57.86%)	1,132 (45.07%)	1,091 (44.62%)
Subtotal	1,649	1,364	1,503	1,871	2,686	2,512	2,445

This table reports the descriptive statistics for daily transactions during each two-year period from 1999 to 2012. Panel A reports the average number of daily trades (No. of Trades), the average number of distinct bonds traded per day (No. of Bonds), the average number of distinct issuers whose bonds were traded per day (No. of Issuers), and the average total par value traded per day (Par Value Traded). Panel B reports the average number of daily transactions by trader type: the number of customer purchases from dealers (Customer Purchase), the number of customer sales to dealers (Customer Sale), and the number of trades between dealers (Inter-Dealer Trades). The proportion of each trader type group is reported in the parentheses. Panels C and D report the average number and average total par value of transactions per day by trade size. Trade size groups are defined in Appendix C. The proportion of each trade size group is reported in the parentheses.

Table 3 Changes in Dealer Markups and Price Dispersion

Panel A: Dealer Net Markups								
Sample	Trade Size	No. Obs.	Pre-RTRS (bps)	No. Obs.	Post-RTRS (bps)	Change (bps)	Change (%)	P-Value
All Bonds	All	449,534	209.30	443,907	179.38	-29.92	-14.29%	<.001
	0-25K	251,620	243.69	245,869	214.78	-28.90	-11.86%	<.001
	25-100K	140,981	190.98	138,885	160.27	-30.71	-16.08%	<.001
	100-200K	21,557	143.54	22,227	110.54	-33.00	-22.99%	<.001
	200-500K	19,119	105.26	19,842	79.04	-26.23	-24.92%	<.001
	500-1,000K	7,016	64.93	7,086	45.17	-19.77	-30.44%	<.001
	1,000K+	9,241	30.54	9,998	21.48	-9.05	-29.65%	<.001
Same Issuers	All	444,240	209.70	430,926	179.22	-30.48	-14.53%	<.001
	0-25K	248,506	244.12	239,775	214.40	-29.72	-12.18%	<.001
	25-100K	139,690	191.34	134,498	159.73	-31.61	-16.52%	<.001
	100-200K	21,340	143.83	21,455	109.98	-33.85	-23.54%	<.001
	200-500K	18,822	105.26	18,989	78.05	-27.21	-25.85%	<.001
	500-1,000K	6,850	64.81	6,735	44.42	-20.39	-31.46%	<.001
	1,000K+	9,032	29.80	9,474	21.12	-8.68	-29.13%	<.001
Same Bonds	All	370,687	223.56	356,733	183.79	-39.77	-17.79%	<.001
	0-25K	213,289	254.70	208,401	215.76	-38.94	-15.29%	<.001
	25-100K	115,900	204.53	108,768	160.75	-43.78	-21.40%	<.001
	100-200K	16,709	157.59	16,388	109.44	-48.14	-30.55%	<.001
	200-500K	13,932	118.20	13,447	75.93	-42.27	-35.76%	<.001
	500-1,000K	4,734	75.54	4,387	43.14	-32.40	-42.89%	<.001
	1,000K+	6,123	33.26	5,342	20.38	-12.88	-38.73%	<.001

Table 3 Change in Dealer Markups and Price Dispersion (Continued)

Panel B: Price Dispersion								
Sample	Trade Size	No. Obs.	Pre-RTRS	No. Obs.	Post-RTRS	Change (\$)	Change (%)	P-Value
All Bonds	All	604,637	0.76	555,233	0.65	-0.11	-14.01%	<.001
	0-25K	339,100	0.93	300,683	0.83	-0.10	-10.72%	<.001
	25-100K	185,491	0.64	176,406	0.53	-0.11	-17.42%	<.001
	100-200K	29,561	0.41	29,679	0.32	-0.09	-21.73%	<.001
	200-500K	28,796	0.30	27,896	0.25	-0.06	-18.89%	<.001
	500-1,000K	9,757	0.20	9,431	0.16	-0.04	-19.90%	<.001
	1,000K+	11,932	0.14	11,138	0.11	-0.02	-16.87%	<.001
Same Issuers	All	598,577	0.76	537,494	0.65	-0.10	-13.76%	<.001
	0-25K	335,619	0.93	292,532	0.83	-0.10	-10.54%	<.001
	25-100K	183,969	0.64	170,331	0.53	-0.11	-17.48%	<.001
	100-200K	29,304	0.41	28,583	0.32	-0.09	-21.95%	<.001
	200-500K	28,385	0.30	26,611	0.25	-0.06	-19.30%	<.001
	500-1,000K	9,539	0.20	8,906	0.16	-0.04	-21.34%	<.001
	1,000K+	11,761	0.14	10,531	0.11	-0.02	-16.33%	<.001
Same Bonds	All	522,266	0.79	444,302	0.69	-0.11	-13.28%	<.001
	0-25K	299,146	0.95	252,157	0.86	-0.10	-10.05%	<.001
	25-100K	158,810	0.68	137,787	0.55	-0.13	-19.25%	<.001
	100-200K	24,631	0.43	22,180	0.33	-0.11	-25.10%	<.001
	200-500K	23,076	0.32	19,316	0.25	-0.08	-23.41%	<.001
	500-1,000K	7,387	0.22	6,065	0.16	-0.06	-27.48%	<.001
	1,000K+	9,216	0.15	6,797	0.12	-0.03	-19.25%	<.001

This table reports the changes in average dealer net markups (bps) and average daily price dispersion (\$) between the pre-RTRS (2004/03-2004/12) and post-RTRS (2005/03-2005/12) periods. In Panel A, dealer net markups are defined as the ratio of the difference between the par-weighted average sale price and the dealer's purchase price to the purchase price, adjusted for the maturity-matched muni index returns. In Panel B, daily price dispersion is defined as the standard deviation of intra-day trading prices for each bond with a minimum of two trades on any given trading day. Changes are reported for three bond samples. The "All Bonds" sample includes all bond. The "Same Issuers" subsample includes bonds issued by the same set of issuers in both time periods. The "Same Bonds" subsample includes bonds that are traded in both time periods. Within each bond sample, changes are reported for each trade size group as defined in Appendix C.

Table 4 Pooled Regression Analysis of Dealer Markups

Variables	0-25K	25-100K	100-200K	200-500K	500-1,000K	1,000K+
RealTime	-16.59*** (0.44)	-13.79*** (0.54)	-14.91*** (1.40)	-12.84*** (1.42)	-10.51*** (2.10)	-4.09*** (1.13)
No. Interdealer (Ln)	90.95*** (0.84)	51.21*** (0.76)	38.49*** (1.57)	32.46*** (1.40)	36.04*** (2.46)	23.69*** (1.63)
RealTime*No. Interdealer	-19.04*** (1.05)	-14.47*** (0.94)	-11.37*** (1.93)	-10.32*** (1.70)	-15.95*** (2.87)	-11.68*** (1.87)
Trade Size (Ln)	-32.95*** (0.33)	-35.98*** (0.48)	-25.63*** (2.33)	-27.95*** (1.41)	-35.73*** (3.47)	-10.79*** (0.65)
AAA	-13.53*** (1.45)	0.34 (1.51)	2.30 (3.07)	0.15 (2.98)	-3.98 (4.01)	0.64 (2.19)
AA	0.88 (1.01)	15.33*** (0.98)	20.19*** (2.05)	14.84*** (2.00)	3.70 (3.01)	-0.54 (1.59)
A-BBB	3.61*** (1.05)	15.74*** (1.08)	23.38*** (2.36)	19.62*** (2.46)	12.45*** (3.84)	4.22** (2.01)
BB-B	29.07*** (4.12)	39.50*** (4.99)	44.24*** (10.90)	51.24*** (12.68)	20.44 (14.41)	11.28 (8.37)
CCC-D	72.83*** (13.08)	51.81*** (19.68)	35.01 (35.14)	45.96 (30.99)	16.68 (39.54)	-23.16 (21.95)
Age (Ln)	-14.09*** (0.77)	-16.04*** (0.69)	-8.68*** (1.38)	-5.10*** (1.29)	0.41 (1.90)	0.33 (1.09)
Maturity (Ln)	75.84*** (0.76)	72.40*** (0.71)	62.78*** (1.45)	46.97*** (1.37)	24.00*** (2.08)	8.54*** (1.17)
Coupon (Ln)	-46.32*** (0.88)	-62.51*** (0.86)	-72.68*** (1.93)	-75.80*** (2.25)	-75.16*** (4.64)	-43.32*** (4.03)
Issue Size (Ln)	-2.80*** (0.35)	-3.80*** (0.36)	-4.38*** (0.77)	-4.11*** (0.74)	-3.76*** (1.10)	-1.87*** (0.65)
Treasury Yield (t-1)	-1.34*** (0.42)	1.76*** (0.42)	3.47*** (0.96)	3.71*** (0.92)	7.07*** (1.38)	3.74*** (0.82)
Treasury Yield Volatility	2.29*** (0.05)	1.75*** (0.05)	1.17*** (0.12)	0.83*** (0.11)	0.71*** (0.17)	0.37*** (0.10)
Intercept	441.84*** (4.38)	496.54*** (5.92)	405.00*** (28.42)	459.46*** (18.46)	585.88*** (47.95)	220.94*** (11.97)
Other Control Variables Included						
No. Obs.	497,489	279,866	43,784	38,961	14,102	19,239
Adj. R-squared	0.33	0.45	0.51	0.49	0.49	0.42
Fixed Effect	Issuer	Issuer	Issuer	Issuer	Issuer	Issuer
SE Cluster	Bond	Bond	Bond	Bond	Bond	Bond

This table reports the pooled regression results on changes in dealer markups by trade size groups. The dependent variable is dealer net markup (bps), defined as the ratio of the difference between the par-weighted average sale price and the dealer's purchase price to the purchase price, adjusted for the maturity-matched muni index returns. RealTime is an indicator variable that equals 1 if the trade occurs after the implementation of RTRS (2005/03-2005/12), and 0 otherwise (2004/03-2004/12). The control variables include both bond/issuer characteristics and general market conditions. All control variables are defined in Appendix C. Issuer fixed effects are included and standard errors are clustered at the bond level. Statistical significance of 1 percent, 5 percent, and 10 percent are indicated by ***, **, and *, respectively. The full version of this regression table is provided in the online appendix.

Table 5 Diff-in-Diff Regression Analysis of Dealer Markups

Variables	0-25K	25-100K	100-200K	200-500K	500-1,000K	1,000K+
Muni	1.05 (4.05)	48.29*** (4.60)	87.30*** (5.93)	51.23*** (4.92)	7.76* (4.64)	-18.05*** (2.25)
RealTime	-7.23* (3.80)	1.14 (3.31)	0.23 (3.30)	-0.55 (2.30)	-11.37*** (2.19)	-2.12** (1.07)
RealTime*Muni	-9.87** (3.93)	-21.68*** (3.45)	-28.78*** (3.90)	-25.55*** (3.11)	-8.22** (3.33)	-5.90*** (1.63)
No. Interdealer (Ln)	40.83*** (2.61)	51.75*** (3.17)	39.14*** (5.38)	20.95*** (4.03)	8.79** (3.52)	2.69 (1.83)
RealTime*No. Interdealer	7.20** (3.20)	-6.33* (3.74)	-0.87 (6.71)	5.42 (5.16)	7.24 (4.87)	1.04 (2.60)
Muni*Interdealer	53.04*** (2.90)	6.05* (3.40)	9.84* (5.96)	24.95*** (4.77)	43.27*** (4.74)	33.03*** (3.05)
RealTime*Muni* No. Interdealer	-22.72*** (3.61)	-4.66 (4.06)	-6.64 (7.52)	-13.91** (6.13)	-21.23*** (6.45)	-16.44*** (3.96)
Trade Size (Ln)	-22.70*** (0.55)	-40.48*** (0.79)	-30.78*** (3.30)	-33.40*** (2.03)	-25.72*** (3.11)	-3.86*** (0.44)
AAA	-26.09*** (1.81)	-25.46*** (1.65)	-32.06*** (2.88)	-20.94*** (2.55)	-14.96*** (2.65)	-5.35*** (1.40)
AA	-12.64*** (1.27)	-12.17*** (1.10)	-13.27*** (2.00)	-11.18*** (1.93)	-8.48*** (2.08)	-6.45*** (1.10)
Age (Ln)	-11.21*** (1.21)	-19.72*** (1.06)	-15.91*** (1.65)	-8.44*** (1.49)	6.81*** (1.64)	7.86*** (1.05)
Maturity (Ln)	97.68*** (2.08)	83.19*** (1.89)	51.00*** (2.36)	35.12*** (1.85)	17.70*** (1.55)	12.43*** (0.95)
Coupon (Ln)	-27.48*** (2.87)	-56.85*** (2.45)	-74.05*** (3.03)	-73.62*** (3.19)	-71.04*** (4.37)	-29.05*** (3.49)
Issue Size (Ln)	-7.21*** (0.55)	-6.25*** (0.48)	-3.48*** (0.77)	-4.42*** (0.73)	-4.22*** (0.73)	-6.04*** (0.40)
Treasury Yield (t-1)	5.34*** (1.21)	-1.30 (1.18)	-1.19 (2.02)	-4.67*** (1.57)	2.67* (1.51)	-0.03 (0.72)
Treasury Volatility	1.90*** (0.08)	1.69*** (0.09)	0.84*** (0.19)	0.76*** (0.17)	0.29* (0.17)	0.13 (0.09)
Intercept	258.69*** (10.05)	523.66*** (11.93)	486.10*** (40.70)	579.34*** (27.55)	481.08*** (44.02)	129.47*** (9.39)
Other Control Variables Included						
No. Obs.	305,203	144,875	21,215	19,739	9,945	22,964
Adj. R-squared	0.26	0.34	0.45	0.45	0.40	0.18
SE Cluster	Bond	Bond	Bond	Bond	Bond	Bond

This table reports the diff-in-diff analysis results on changes in dealer markups using a subset of corporate bonds as the control group. The dependent variable is dealer net markup (bps), defined as the ratio of the difference between the par-weighted average sale price and the dealer's purchase price to the purchase price and adjusted for the maturity-matched muni or corporate index returns. RealTime is an indicator variable that equals 1 if the trade occurs after the implementation of RTRS (2005/03-2005/12), and 0 otherwise (2004/03-2004/12). Muni is an indicator variable that equals 1 for a municipal bond and 0 for a corporate bond. RealTime*Muni is the interaction term between RealTime and Muni. Other interaction terms are similarly defined. All control variables are defined in Appendix C. Standard errors are clustered at the bond level. Statistical significance of 1 percent, 5 percent, and 10 percent are indicated by ***, **, and *, respectively. The full version of this table is provided in the online appendix.

Table 6 Decomposition of Dealer Markups: Intermediation Costs vs. Bargaining Power

Variables	0-25K	25-100K	100-200K	200-500K	500-1,000K	1,000K+
Intermediation Costs	163.60*** (0.43)	117.09*** (0.50)	65.13*** (1.22)	4.29*** (0.67)	-9.06*** (0.55)	-12.02*** (0.35)
RealTime	-12.99*** (0.58)	-10.35*** (0.70)	-16.78*** (1.64)	2.52*** (0.83)	4.84*** (0.80)	3.87*** (0.49)
Trade Size (Ln)	-19.10*** (0.51)	-36.94*** (0.77)	-18.36*** (3.11)	-2.65*** (0.92)	-6.33*** (1.49)	0.14 (0.31)
Other Control Variables Included						
Bargaining Power Gain	74.08*** (0.41)	67.33*** (0.48)	73.24*** (1.19)	98.52*** (0.92)	72.89*** (0.99)	40.28*** (0.51)
RealTime	-4.54*** (0.54)	-7.17*** (0.63)	-6.32*** (1.44)	-25.03*** (1.07)	-23.91*** (1.20)	-11.52*** (0.63)
Trade Size (Ln)	-18.22*** (0.65)	-1.66 (1.17)	-24.15*** (4.62)	-38.80*** (2.00)	-75.72*** (4.48)	-41.97*** (1.33)
Symmetric Error	93.05*** (0.25)	79.84*** (0.30)	64.23*** (0.77)	22.66*** (0.52)	15.41*** (0.36)	14.62*** (0.21)
RealTime	-10.85*** (0.34)	-11.20*** (0.39)	-14.91*** (0.96)	-4.45*** (0.51)	-1.08** (0.44)	-0.39 (0.27)
Trade Size (Ln)	-13.56*** (0.34)	-15.76*** (0.64)	-36.88*** (3.73)	-10.24*** (3.49)	-3.34 (6.71)	-1.50 (1.25)
-2 Log Likelihood	6,099,616	3,347,336	513,274	440,189	149,845	188,362
No. Obs.	497,489	279,866	43,784	38,961	14,102	19,239

This table reports the maximum likelihood estimation of Eq. (3) that decomposes dealer net markup into the Intermediation Costs and Bargaining Power Gain components. The dependent variable is dealer net markup (bps), defined as the ratio of the difference between the par-weighted average sale price and the dealer's purchase price to the purchase price and adjusted for the maturity-matched muni index returns. Intermediation Costs is the constant term θ_0 in Eq. (4). Bargaining Power Gain is the constant term α_0 in the conditional mean of the one-sided error term as specified in Eq. (5). Symmetric Error is the constant term in the standard deviation of symmetric error term as specified in Eq. (A3). RealTime is an indicator variable that equals 1 if the trade occurs after the implementation of RTRS (2005/03-2005/12) and 0 otherwise (2004/03-2004/12). All control variables are demeaned and defined in Appendix C. Statistical significance of 1 percent, 5 percent, and 10 percent are indicated by ***, **, and *, respectively. The full version of this table is provided in the online appendix.

Table 7 Bond Characteristics by Pre-RTRS Trading Volume Groups

Group	N	Trading Volume	Markup (bps)	Price Dispersion (\$)	Offering Amount (\$)	Bond Age (years)	Coupon (%)	Insured
0	74,606	0.00%			6,549,330	5.01	5.40	46.26%
1	18,609	0.18%	232.16	1.02	10,252,508	5.70	5.25	49.94%
2	18,609	0.58%	217.48	0.86	7,758,692	5.27	5.07	56.14%
3	18,609	1.22%	208.56	0.78	8,356,394	4.83	4.95	58.55%
4	18,609	2.66%	191.28	0.65	7,927,526	4.42	4.81	58.96%
5	18,609	11.03%	156.79	0.52	8,953,549	3.74	4.77	58.14%
All	167,651	1.74%	201.12	0.75	7,715,038	4.89	5.16	51.86%

Group	N	AAA	AA	A-BBB	BB-D	Unrated	Guaranteed	Letter/Line
0	74,606	9.26%	21.11%	13.14%	0.57%	55.92%	16.93%	0.56%
1	18,609	7.95%	19.17%	14.03%	0.74%	58.12%	13.64%	0.52%
2	18,609	7.23%	19.86%	15.20%	0.55%	57.17%	14.99%	0.51%
3	18,609	6.84%	19.69%	16.21%	0.70%	56.55%	16.13%	0.47%
4	18,609	6.06%	18.73%	15.04%	0.81%	59.36%	18.25%	0.29%
5	18,609	7.08%	22.07%	15.88%	0.96%	54.01%	20.58%	0.28%
All	167,651	8.02%	20.44%	14.32%	0.67%	56.54%	16.81%	0.48%

This table reports mean statistics for six bond groups sorted by trading volume. Trading volume is defined as par value traded scaled by the bond's offering amount. Bonds are first sorted into six groups based on their average monthly trading volume in 2003. Group 0 includes bonds with no trades recorded in the MSRB transaction database for 2003. Conditional on observing at least 1 trade in 2003, the remaining bonds are sorted into 5 quintile groups (Groups 1 to 5) based on their average monthly trading volume in 2003 in ascending order. The mean statistics of key bond characteristics are computed for each trading volume group during the pre-RTRS period (2004/03-2004/12). Markup is defined as the ratio of the difference between the par-weighted average sale price and the dealer's purchase price to the purchase price and adjusted for the maturity-matched muni index returns. Price Dispersion is defined as the standard deviation of intra-day trading prices for each bond with a minimum of two trades on any given trading day. All other bond characteristics are defined in Appendix C.

Table 8 Diff-in-Diff Regression Analysis of Trading Volume

Variables	Group 0	Group 1	Group 2	Group 3	Group 4	Group 5
Muni	0.07 (0.12)	-0.08 (0.16)	-0.60*** (0.19)	-1.56*** (0.25)	-2.60*** (0.25)	-6.75*** (0.48)
RealTime	0.03 (0.12)	-0.09 (0.10)	-0.12 (0.12)	-0.58*** (0.15)	-0.59*** (0.23)	-1.04** (0.45)
RealTime*Muni	0.24** (0.12)	0.36*** (0.11)	0.25* (0.13)	0.62*** (0.16)	0.43* (0.22)	0.78* (0.43)
AAA	0.20*** (0.05)	0.07 (0.07)	0.15** (0.07)	0.12* (0.07)	0.15 (0.09)	-0.15 (0.16)
AA	0.12*** (0.04)	0.06 (0.06)	0.12** (0.05)	0.02 (0.05)	-0.03 (0.07)	-0.27** (0.11)
Age (Ln)	-0.44*** (0.04)	-0.47*** (0.06)	-0.53*** (0.07)	-0.53*** (0.07)	-0.51*** (0.08)	-0.20 (0.14)
Maturity (Ln)	-0.50*** (0.06)	-0.46*** (0.09)	-0.42*** (0.09)	-0.53*** (0.09)	-0.51*** (0.10)	0.14 (0.21)
Coupon (Ln)	0.06 (0.05)	-0.04 (0.11)	-0.38** (0.15)	-0.50*** (0.16)	-0.96*** (0.13)	-1.87*** (0.16)
Issue Size (Ln)	-0.10*** (0.02)	-0.00 (0.02)	0.09*** (0.02)	0.12*** (0.02)	0.18*** (0.02)	0.41*** (0.05)
VIX Change (t-1)	-0.02*** (0.01)	-0.01 (0.01)	0.01 (0.01)	-0.01 (0.01)	-0.02* (0.01)	-0.06*** (0.02)
Libor Rate Change (t-1)	-1.39*** (0.23)	-1.54*** (0.28)	-1.83*** (0.27)	-2.52*** (0.28)	-2.62*** (0.33)	-4.24*** (0.63)
Treasury Yield (t-1)	0.32*** (0.06)	0.25*** (0.07)	0.16** (0.07)	0.43*** (0.07)	0.41*** (0.09)	0.66*** (0.16)
Treasury Yield Volatility	0.01 (0.01)	0.00 (0.01)	-0.00 (0.01)	0.01 (0.01)	0.00 (0.01)	0.01 (0.02)
Intercept	1.03*** (0.23)	2.18*** (0.31)	4.07*** (0.41)	4.66*** (0.44)	6.52*** (0.43)	10.02*** (0.82)
Other Control Variables Included						
Observations	623,740	158,980	158,960	158,960	158,960	158,940
Adj. R-squared	0.00	0.00	0.00	0.01	0.01	0.03
SE Cluster	Bond	Bond	Bond	Bond	Bond	Bond

This table reports the diff-in-diff analysis on the changes in trading volume using a subset of corporate bonds disseminated in the first two phases of TRACE as the control group. Bonds are sorted into six groups based on their average monthly trading volume in 2003 as in Table 7 and regression is estimated separately for each trading volume group. The dependent variable is a bond's monthly trading volume defined as par value traded scaled by the bond's issue amount. RealTime is an indicator variable that equals 1 if the trade occurs after the implementation of RTRS (2005/03-2005/12), and 0 otherwise (2004/03-2004/12). Muni is an indicator variable that equals 1 for a municipal bond, and 0 for a corporate bond. RealTime*Muni is the interaction term between Muni and RealTime indicators. All control variables are defined in Appendix C. Standard errors are clustered at the bond level. Statistical significance of 1 percent, 5 percent, and 10 percent are indicated by ***, **, and *, respectively. The full version of this table is provided in the online appendix.

Table 9 Panel A. Diff-in-Diff Regression Analysis of Dealer Capital Commitment (Daily)

Variables	Group 0	Group 1	Group 2	Group 3	Group 4	Group 5
Muni	0.12*** (0.03)	-1.53*** (0.26)	-1.71*** (0.41)	-0.86* (0.47)	-2.44*** (0.48)	-6.86*** (0.62)
RealTime	0.17*** (0.05)	-0.63*** (0.10)	-1.09*** (0.16)	-1.70*** (0.21)	-2.39*** (0.21)	-2.51*** (0.23)
RealTime*Muni	-0.11** (0.05)	0.39*** (0.10)	0.52*** (0.14)	0.97*** (0.19)	1.46*** (0.19)	1.94*** (0.21)
AAA	-0.02 (0.01)	-0.15** (0.07)	-0.16* (0.09)	-0.15 (0.12)	-0.21* (0.13)	-0.60*** (0.12)
AA	0.01 (0.01)	-0.02 (0.06)	-0.03 (0.08)	0.07 (0.10)	-0.15 (0.10)	-0.41*** (0.10)
Age (Ln)	-0.00 (0.01)	0.15** (0.07)	0.29*** (0.10)	0.31*** (0.11)	0.29** (0.11)	0.22* (0.12)
Maturity (Ln)	0.02 (0.01)	0.33*** (0.07)	0.71*** (0.12)	1.08*** (0.14)	1.17*** (0.13)	1.49*** (0.15)
Coupon (Ln)	-0.11*** (0.01)	-0.51*** (0.19)	-1.35*** (0.25)	-1.10*** (0.20)	-1.70*** (0.12)	-1.91*** (0.10)
Issue Size (Ln)	0.07*** (0.01)	0.57*** (0.03)	1.56*** (0.04)	2.33*** (0.06)	2.32*** (0.06)	1.87*** (0.06)
VIX Change (t-1)	-0.00 (0.00)	0.01** (0.00)	0.05*** (0.01)	0.07*** (0.01)	0.09*** (0.01)	0.05*** (0.01)
Libor Rate Change (t-1)	-0.10*** (0.03)	-2.12*** (0.14)	-4.50*** (0.21)	-6.63*** (0.25)	-6.92*** (0.24)	-5.64*** (0.24)
Treasury Yield (t-1)	0.03*** (0.01)	0.21*** (0.05)	0.40*** (0.07)	0.96*** (0.09)	0.97*** (0.09)	0.81*** (0.09)
Treasury Volatility (t-1)	0.00** (0.00)	0.01* (0.01)	0.02** (0.01)	0.03*** (0.01)	0.04*** (0.01)	0.01 (0.01)
Intercept	-0.08 (0.06)	0.93* (0.49)	1.39** (0.69)	-2.62*** (0.72)	-0.68 (0.75)	3.96*** (0.95)
Other Control Variables Included						
Observations	623,740	158,980	158,960	158,960	158,960	158,940
Adj. R-squared	0.07	0.13	0.30	0.45	0.53	0.66
SE Cluster	Bond	Bond	Bond	Bond	Bond	Bond

Table 9 Panel B. Diff-in-Diff Regression Analysis of Dealer Capital Commitment (Weekly)

Variables	Group 0	Group 1	Group 2	Group 3	Group 4	Group 5
Muni	0.33*** (0.08)	-6.69*** (0.62)	-7.17*** (0.78)	-10.93*** (0.90)	-13.64*** (0.93)	-15.03*** (1.19)
RealTime	0.42*** (0.10)	-2.74*** (0.37)	-2.93*** (0.44)	-3.17*** (0.47)	-3.27*** (0.50)	-1.73*** (0.43)
RealTime*Muni	-0.22** (0.10)	1.85*** (0.36)	1.67*** (0.43)	1.98*** (0.46)	2.07*** (0.49)	1.08*** (0.42)
AAA	-0.03 (0.03)	-0.59*** (0.16)	-0.39* (0.20)	-0.45* (0.24)	-0.88*** (0.24)	-1.76*** (0.25)
AA	0.04 (0.03)	-0.06 (0.14)	0.03 (0.16)	0.21 (0.18)	-0.38** (0.18)	-1.06*** (0.19)
Age (Ln)	-0.14*** (0.03)	0.00 (0.15)	0.12 (0.19)	-0.07 (0.22)	0.42* (0.22)	0.88*** (0.23)
Maturity (Ln)	-0.00 (0.04)	1.20*** (0.22)	2.30*** (0.27)	3.41*** (0.28)	4.25*** (0.28)	5.19*** (0.29)
Coupon (Ln)	-0.26*** (0.05)	-0.75* (0.43)	-1.51*** (0.34)	-2.85*** (0.44)	-3.32*** (0.30)	-4.56*** (0.25)
Issue Size (Ln)	0.10*** (0.01)	1.45*** (0.07)	3.46*** (0.08)	4.28*** (0.09)	4.16*** (0.09)	3.43*** (0.10)
VIX Change (t-1)	-0.01** (0.00)	0.00 (0.02)	0.08*** (0.02)	0.06*** (0.02)	0.06*** (0.02)	-0.00 (0.02)
Libor Rate Change (t-1)	-0.07 (0.08)	-4.40*** (0.48)	-6.11*** (0.56)	-7.44*** (0.59)	-7.46*** (0.59)	-6.34*** (0.58)
Treasury Yield (t-1)	-0.00 (0.03)	-0.08 (0.15)	-0.08 (0.18)	0.33* (0.19)	0.41** (0.19)	0.64*** (0.19)
Treasury Volatility (t-1)	0.01*** (0.00)	0.11*** (0.02)	0.14*** (0.02)	0.21*** (0.02)	0.17*** (0.02)	0.12*** (0.02)
Intercept	0.66*** (0.15)	8.88*** (1.36)	7.94*** (1.27)	8.30*** (1.30)	7.84*** (1.32)	8.36*** (1.63)
Other Control Variables Included						
Observations	623,740	158,980	158,960	158,960	158,960	158,940
Adj. R-squared	0.05	0.10	0.21	0.28	0.31	0.32
SE Cluster	Bond	Bond	Bond	Bond	Bond	Bond

This table reports the diff-in-diff analysis on the changes in daily (Panel A) or weekly (Panel B) dealer capital commitment using a subset of corporate bonds disseminated in the first two phases of TRACE as the control group. Bonds are sorted into six groups based on their average monthly trading volume in 2003 as in Table 7 and regression is estimated separately for each trading volume group. The dependent variable is the monthly average of aggregate daily (or weekly) dealer capital commitment for each bond, calculated as the absolute value of the dealers' aggregate buy volume less the dealers' aggregate sell volume during the day (or week), scaled by the total daily (or weekly) trading volume for the bond. RealTime is an indicator variable that equals 1 if the trade occurs after the implementation of RTRS (2005/03-2005/12), and 0 otherwise (2004/03-2004/12). Muni is an indicator variable that equals 1 for a municipal bond, and 0 for a corporate bond. RealTime*Muni is the interaction term between the two indicators. All control variables are defined in Appendix C. Standard errors are clustered on the bond level. Statistical significance of 1 percent, 5 percent, and 10 percent are indicated by ***, **, and *, respectively. The full version of this table is reported in the online appendix.

Table 10 Length of Intermediation Chains Analysis by Trading Volume Groups

Panel A: Number of Days/Sequence							
Group	Pre-RTRS		Post-RTRS		Change (#days)	Change (%)	P-Value
	No. Sequences	No. Days	No. Sequences	No. Days			
0	19,664	4.20	28,731	3.92	-0.28	-6.78%	<.001
1	35,081	3.66	33,800	3.55	-0.12	-3.22%	0.005
2	59,736	3.21	56,127	3.25	0.04	1.25%	0.18
3	75,789	3.07	70,899	3.16	0.10	3.15%	<.001
4	66,706	3.21	64,676	3.30	0.08	2.62%	0.002
5	53,697	3.37	54,843	3.47	0.09	2.77%	0.003
All	310,673	3.23	309,076	3.27	0.04	1.09%	<.001

Panel B: Number of Interdealer Trades/Sequence							
Group	Pre-RTRS		Post-RTRS		Change (#trades)	Change (%)	P-Value
	No. Sequences	No. Interdealer Trades	No. Sequences	No. Interdealer Trades			
0	19,664	0.44	28,731	0.45	0.01	2.49%	0.219
1	35,081	0.35	33,800	0.41	0.05	14.87%	<.001
2	59,736	0.34	56,127	0.39	0.05	16.18%	<.001
3	75,789	0.33	70,899	0.38	0.05	14.69%	<.001
4	66,706	0.37	64,676	0.42	0.05	13.52%	<.001
5	53,697	0.42	54,843	0.47	0.06	13.48%	<.001
All	310,673	0.36	309,076	0.41	0.05	13.60%	<.001

This table reports the changes in the length of intermediation chains between the pre-RTRS (2004/03-2004/12) and the post-RTRS (2005/03-2005/12) periods. Number of Days per sequence measures the number of days for a roundtrip trade from the dealer's first purchase to the last sale. Number of Interdealer Trades per sequence measures the number of interdealer trades involved in each roundtrip trade. Bonds are sorted into six groups based on their average monthly trading volume in 2003 as in Table 7. For each trading volume group, the average Number of Days (Panel A) and the average Number of Interdealer Trades (Panel B) are reported for the pre-RTRS and the post RTRS periods, respectively. Change is measured as the difference between the pre-RTRS mean and the post-RTRS mean.

Table 11 Changes in Trade Allocation by Match Groups

Trade Size	Match Group	Pre-RTRS		Post-RTRS		Chang (%)
		No. Trades	Percent	No. Trades	Percent	
0-25K	Pre-arranged	17,272	6.86%	7,909	3.22%	-54.21%
	Same-day	61,539	24.46%	58,166	23.66%	-5.48%
	Roundtrip	147,268	58.53%	156,461	63.64%	6.24%
	FIFO	25,541	10.15%	23,333	9.49%	-8.64%
	Subtotal	251,620	100.00%	245,869	100.00%	-2.29%
25-100K	Pre-arranged	10,061	7.14%	4,467	3.22%	-55.60%
	Same-day	21,635	15.35%	24,127	17.37%	11.52%
	Roundtrip	86,193	61.14%	91,025	65.54%	5.61%
	FIFO	23,092	16.38%	19,266	13.87%	-16.57%
	Subtotal	140,981	100.00%	138,885	100.00%	-1.49%
100-200K	Pre-arranged	1,538	7.13%	724	3.26%	-52.93%
	Same-day	2,964	13.75%	3,817	17.17%	28.78%
	Roundtrip	12,724	59.02%	13,930	62.67%	9.48%
	FIFO	4,331	20.09%	3,756	16.90%	-13.28%
	Subtotal	21,557	100.00%	22,227	100.00%	3.11%
200-500K	Pre-arranged	1,525	7.98%	661	3.33%	-56.66%
	Same-day	3,095	16.19%	3,886	19.58%	25.56%
	Roundtrip	10,482	54.83%	11,806	59.50%	12.63%
	FIFO	4,017	21.01%	3,489	17.58%	-13.14%
	Subtotal	19,119	100.00%	19,842	100.00%	3.78%
500-1,000K	Pre-arranged	687	9.79%	223	3.15%	-67.54%
	Same-day	1,503	21.42%	1,939	27.36%	29.01%
	Roundtrip	3,431	48.90%	3,744	52.84%	9.12%
	FIFO	1,395	19.88%	1,180	16.65%	-15.41%
	Subtotal	7,016	100.00%	7,086	100.00%	1.00%
1,000K+	Pre-arranged	1,154	12.49%	367	3.67%	-68.20%
	Same-day	2,321	25.12%	3,110	31.11%	33.99%
	Roundtrip	4,122	44.61%	5,036	50.37%	22.17%
	FIFO	1,644	17.79%	1,485	14.85%	-9.67%
	Subtotal	9,241	100.00%	9,998	100.00%	8.19%
All	Pre-arranged	32,237	7.17%	14,351	3.23%	-55.48%
	Same-day	93,057	20.70%	95,045	21.41%	2.14%
	Roundtrip	264,220	58.78%	282,002	63.53%	6.73%
	FIFO	60,020	13.35%	52,509	11.83%	-12.51%
	Total	449,534	100.00%	443,907	100.00%	-1.25%

This table reports the changes in the proportions of four different types of matched trades between the pre-RTRS (2004/03-2004/12) and post-RTRS (2005/03-2005/12) periods. For each trade size group, the number, proportion, and percentage change of four types of matched trades are reported. Pre-arranged matches comprise trading pairs where the time stamp, par value, and CUSIP are all identical for a dealer buy and a dealer sell. Same-day matches comprise

trading pairs where a dealer buy is followed by a dealer sell for the same bond and the same par amount on the same day, with no intervening purchases from customers. Roundtrip matches comprise trades where a dealer buy is followed by a series of dealer sells with no intervening purchases from customers. FIFO matches comprise trades matched using the first-in-first-out (FIFO) rule to deal with intervening purchases from customers.

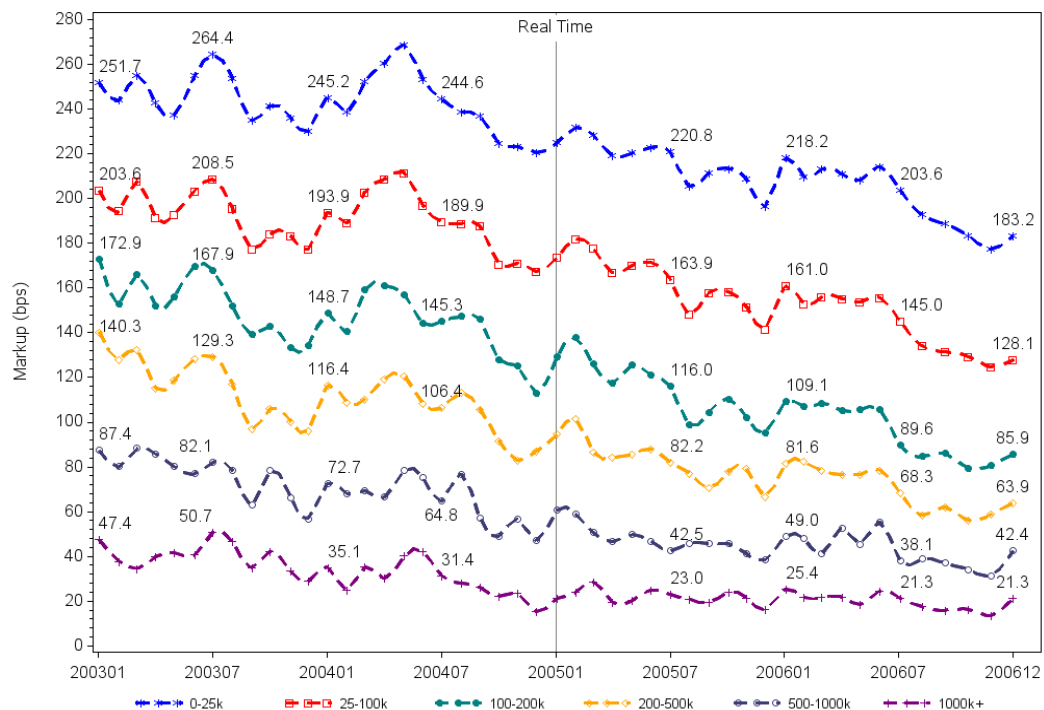


Figure 1

Dealer Markups in the Municipal Bond Market by Trade Size Groups

This figure plots monthly average net markups for each trade size group from 2003/01 to 2006/12. Dealer net markup is defined as the ratio of the difference between the par-weighted average sale price and the dealer's purchase price to the purchase price and adjusted for the maturity-matched muni index returns. Trade size groups are defined in Appendix C.

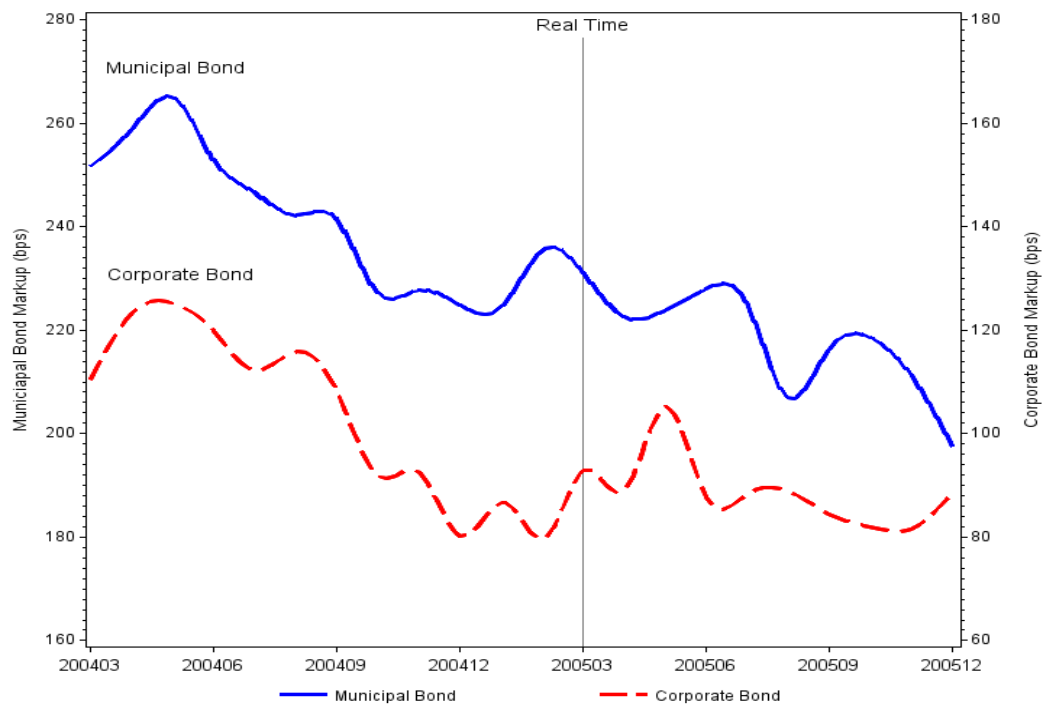


Figure 2

Dealer Markups for Municipal Bonds and the Control Group of Corporate Bonds

This figure plots monthly average dealer net markups of municipal bonds and the control group of corporate bonds disseminated during the first two phases of TRACE from 2004/03 to 2005/12. Dealer net markup is defined as the ratio of the difference between the par-weighted average sale price and the dealer's purchase price to the purchase price and adjusted for the maturity-matched muni or corporate index returns.

Appendix A: History of MSRB Market Transparency Enhancement

MSRB (Municipal Securities Rulemaking Board) is a self-regulatory organization created under the Securities Acts Amendments of 1975 and a Virginia non-stock corporation. It protects investors and municipal entities in the municipal market by making market information as transparent and widely available as possible.

Since January 1994, MSRB started to improve the market transparency and its goal was to make trading information available for investors in a contemporaneous manner. Municipal security price information started to become available from January 1995 in “Interdealer Daily Reports” which included such interdealer market activity information as price and volume of frequently traded securities from the previous day. Trading information for securities traded infrequently was not published. In August 1998, MSRB produced “Combined Daily Reports” which merged trading information from customer and interdealer. The reports offered daily high, low, and average prices for frequently traded municipal securities on a one-day delayed basis. Both customer and interdealer transactions are included in “Combined Daily Report” and infrequently traded securities are still not available in this report. On January 15th, 2000, the MSRB brought out “Daily Transaction Reports” which included CUSIP, issue description, trading price, par value, trading time reported by dealer, and indicator of whether the trade was a sale to a customer, a purchase from a customer, or an interdealer transaction. The report still focused on frequently traded securities on a one-day lag.²⁸

In October 2000, the MSRB introduced “Monthly Comprehensive Reports” that provided information for all transactions from previous month. The report disclosed trade date, CUSIP, issue description, par value, trade time reported by dealer, trading price, the dealer-reported yield, and an indicator of whether the transaction was a sale to customer, a purchase from a customer, or an

²⁸ Four or more interdealer transactions in a single day meet the frequency criterion.

interdealer trade on a one-month lag. In February 2001, the MSRB released “Statistical Patterns in the Municipal Market” which summarized charts and tables based on trading activity, security type, security age, and trade size on a one- or two- month lag. In November 2001, the MSRB introduced “Daily Comprehensive Report” which was based on a two-week lag. On May 16th, 2002, the MSRB shorten the frequency criterion from four transactions to three per day. In August 2002, the MSRB shorten the lag of “Daily Comprehensive Report” to one week. On December 3rd, 2002, the MSRB lowered the frequency criterion to two transactions per day. On June 23rd, 2003, the MSRB presented the enhanced “T+1 Daily Report” which included all transactions with a par value of one million dollars or less²⁹.

On January 31st, 2005, the MSRB presented “Real-Time Transaction Reporting System (RTRS)” which required municipal securities dealers to report most transactions to the MSRB within 15 minutes of the time of trade and the information was made available on EMMA (Electronic Municipal Market Access) website immediately³⁰. The specific items disseminated by this report are CUSIP, par value, trading price, yield, trading date and time, an indicator of whether the trade was a sale to a customer, a purchase from a customer, or an interdealer transaction and some other supplementary information. The RTRS website is available for viewing trade information 24 hours per day and 7 days per week. The report submission starts from 6:00 a.m. until 9:00 p.m. ET on RTRS Business Days³¹. However, RTRS accepts and disseminates any trade report received between the RTRS “window” hours³² of 6:00 a.m. and 9:00 p.m. ET. Transactions executed

²⁹ The enhanced T+1 Daily Report will display exact par value for all trades with a par value of \$1 million and under. It displays “\$1 MM+” for trades that have a par value greater than \$1 million.

³⁰ Unless the trade report contains errors or is subject to exception, transactions reported by dealers during the RTRS Business Day are disseminated within a few minutes after receipt at the designated RTRS Portal. See Rule G-14 for more details, <http://www.msrb.org/Rules-and-Interpretations/MSRB-Rules/General/Rule-G-14.aspx>

³¹ The RTRS “Business Day”, on which dealers are required by Rule G-14 to submit trade reports within 15 minutes of execution, begins at 7:30 a.m. and ends at 6:30 p.m. ET.

³² RTRS “Window” Hours begin at 6:00 a.m. and end at 9:00 p.m. ET.

outside the RTRS Business Day are required to be reported but not disseminated on a real-time basis. Trade reports submitted to RTRS after 9:00 p.m. will be processed and disseminated at 6:00 a.m. the next business day. Transactions with a par value exceeding \$1 million will initially be disseminated with a par value of “1MM+”. The actual par value will be shown in five business days after trade date³³.

This reporting system largely improved the price transparency for the municipal market. This paper uses two sample periods covering transactions recorded in 2004 and 2005. Based on the improvement in transaction information availability after 31st January 2005, we expect a reduction in transaction costs after the implementation of RTRS.

³³ Starting from November 5, 2012, the threshold is raised from \$1 million to \$5 million.

Appendix B: Decomposition of Dealer Markups (Green, Hollifield, and Schürhoff 2007b)

For convenience, we offer a description of the model we use to disaggregate transaction costs into intermediation costs and bargaining power. Consider a customer who wishes to sell some bonds to a dealer. The customer has a reservation value v and will sell only if the price offered by the dealer exceeds the reservation value. The dealer incurs intermediation costs c if she trades with the customer and will do so only if the expected resale price in the future $E(p)$ exceeds the sum of intermediation costs c and purchase price p^* . The participation constraints for the customer and the dealer imply that a transaction would occur between the two parties only if the total trading surplus is positive, or $E(p) - c - v > 0$. In this case, the dealer (with bargaining power ρ) engages in a Nash bargaining game with the customer (with bargaining power $1 - \rho$) that determines the equilibrium purchase price p^* and the dealer markup $p - p^*$.

The first term on the right side of (4) reflects the costs of intermediating the trade, which likely depends on various trade and bond characteristics. For example, it is relatively easier for the dealer to find a counterparty for a bond with larger issue size and higher credit rating. To parameterize *Intermediation Costs*, we use the linear specification:

$$\text{Intermediation Costs} = \theta_0 + \theta_{01} \text{RealTime} + \theta X, \quad (\text{A1})$$

where X is a vector of bond and trade characteristics, θ is the corresponding vector of coefficients, and θ_0 is the constant term. The X vector includes par value traded, coupon rate, time to maturity, issue size, credit rating, redemption provisions, insurance provisions, bond types, and other characteristics. To capture the effect of RTRS on the intermediation costs, we add an indicator variable *RealTime* to the constant term that equals one if the trade occurs after the implementation of RTRS and zero otherwise. The coefficient θ_{01} therefore reflects the change in average intermediation costs after the RTRS.

Bargaining Power Gain in (4) measures the dealer's share of trading surplus, which depends on her bargaining power ρ relative to the customer and the size of total surplus $E(p) - c - v$. If the dealer has all the bargaining power ($\rho = 1$), she would extract all the trading surplus. To estimate the *bargaining power* component of dealer markup, we assume that the customer's reservation value v draws from a distribution with mean equal to the dealer's reservation value $E(p) - c$. Since trades occur only when the surplus is positive, the realized dealer surplus draws from a one-sided distribution. We parameterize this one-sided conditional distribution as an exponential distribution. For empirical estimation, we assume that the conditional mean (and also the conditional standard deviation) of the dealer's surplus is a log-linear function of bond and trade characteristics:

$$E(\text{Bargaining Power Gain} | \text{Positive Trading Surplus}) = (a_0 + a_{01} \text{RealTime}) * e^{aZ}, \quad (\text{A2})$$

where Z is the vector of explanatory variables, a is the vector of coefficients, and a_0 is the constant term. To keep the specification parsimonious, we include in Z only the logarithm of par value traded, which is a known factor affecting the trading costs. Larger size trades are likely carried out by sophisticated institutional investors with better bargaining positions. We add the *RealTime* indicator variable to the constant term and its coefficient a_{01} should capture the impact of RTRS on the dealer's share of trading surplus in a typical transaction.

Finally, we assume that the last error term in (4) is symmetric with mean zero and standard deviation σ . We also parameterize the standard deviation as a log-linear function of explanatory variables Z :

$$\sigma = (b_0 + b_{01} \text{RealTime}) * e^{bZ}. \quad (\text{A3})$$

As before, we add the *RealTime* indicator to the constant term and focus on the coefficient b_{01} to capture the impact of RTRS.

Appendix C: Variable Definition

Panel A: Bond Characteristics

Variable	Description
AAA	Indicator variable equal to 1 if the bond's credit rating is AAA and 0 otherwise. (The lowest available score among S&P, Moody's and Fitch – same for all rating variables below.)
AA	Indicator variable equal to 1 if the bond's credit rating is AA and 0 otherwise.
A-BBB	Indicator variable equal to 1 if the bond's credit rating is between A+ and BBB- and 0 otherwise.
Age	The number of years since the bond was issued.
BB-D	Indicator variable equal to 1 if the bond's credit rating is between BB+ and D- and 0 otherwise.
Callable bond	Indicator variable equal to 1 if the bond is callable and 0 otherwise.
Coupon	The coupon rate of the bond (%).
Coupon zero or missing	Indicator variable equal to 1 if the bond's coupon information is zero or missing and 0 otherwise.
Development authority	Indicator variable equal to 1 if the bond issuer is a development authority and 0 otherwise.
Facility construction	Indicator variable equal to 1 if the bond is issued to finance facility constructions and 0 otherwise.
General obligation bond	Indicator variable equal to 1 if the bond is backed by the credit and taxing power of the issuing jurisdiction rather than the revenue from a given project and 0 otherwise.
Guaranteed bond	Indicator variable equal to 1 if the bond's timely interest and principal payments are guaranteed by a third party and 0 otherwise.
Housing authority	Indicator variable equal to 1 if the bond issuer is a housing authority and 0 otherwise.
Improvement	Indicator variable equal to 1 if the bond is issued to finance facility improvements and 0 otherwise.
Insured bond	Indicator variable equal to 1 if the bond has an insurance policy covering default underwritten by a private insurance company and 0 otherwise.
Issue size	The offering amount of the bond (\$).
Letter or line of credit	Indicator variable equal to 1 if the bond is backed by a letter or line of credit and 0 otherwise.
Maturity	The number of years left until the maturity date.
Pre-refunded bond	Indicator variable equal to 1 if the bond has been pre-refunded and 0 otherwise.
Rated	Indicator variable equal to 1 if the bond's credit rating information is available and 0 otherwise.
Redevelopment	Indicator variable equal to 1 if the issuer is a redevelopment authority and 0 otherwise.
Refunding	Indicator variable equal to 1 if the bond is issued to refund another bond issue and 0 otherwise.
Revenue bond	Indicator variable equal to 1 if the bond finances income-producing projects and is secured by a specified revenue source and 0 otherwise.
Sewer authority	Indicator variable equal to 1 if the issuer is a sewer authority and 0 otherwise.
School bond	Indicator variable equal to 1 if the bond is issued by a city or school district to fund school construction and 0 otherwise.
Taxable bond	Indicator variable equal to 1 if the bond's income is not exempt from federal tax and 0 otherwise.
Tax revenue bond	Indicator variable equal to 1 if the bond will be paid by specific tax revenues and 0 otherwise.
Utility construction	Indicator variable equal to 1 if the bond is issued to finance utility construction and 0 otherwise.

Panel B: Trade Characteristics and Market Conditions

Variable	Description
Bond index return	Monthly index return calculated based on municipal or corporate bond daily market index.
Bond index volatility	Monthly standard deviation of index returns calculated based on municipal or corporate bond daily market index.
Corporate market index	Bank of America's corporate bond daily index return with maturities in 1, 3, 5, 7, 10 and 15+ years.
Libor rate change	Monthly change in 1-month London Interbank Offered Rate (%).
Monthly volume	Total monthly par value traded across all active bonds in the municipal or corporate market (million \$).
Municipal market index	Barclays' municipal bond daily index return with maturities in 1, 3, 5, 7, 10, 15, 20 and 22+ years.
No. Interdealer	The number of inter-dealer transactions within a Roundtrip or FIFO match.
State new issues	Total par value of new bond issuance by the state per day averaged over the last 7 days ending the day prior to the initial dealer purchase transaction. (billion \$)
State order imbalance	Difference in par values between dealer purchase from customers and dealer sale to customers for bonds issued by the state per day averaged over the last 7 days ending the day prior to the initial dealer purchase transaction. (billion \$)
State volume	The total par value of trades per day in bonds issued by the state averaged over the last 7 days ending the day prior to the initial dealer purchase transaction (billion \$).
Trade size	The 0-25k group includes trade size (par value traded) in (\$0, \$25,000]. The 25-100k group includes trade size in (\$25,000, \$100,000]. The 100-200k group includes trade size in (\$100,000, \$200,000]. The 200-500k group includes trade size in (\$200,000, \$500,000]. The 500-1,000k group includes trade size in (\$500,000, \$1,000,000]. The 1,000k+ group includes trade size larger than \$1,000,000.
Transaction frequency	The average number of purchase transactions per day over the minimum of the bond's age, the number of days since the beginning of the sample period, and the previous 365 days, ending the day prior to the initial dealer purchase transaction.
Treasury volatility	The volatility (bps) of maturity matched daily treasury yield over 30 days ending the day prior to the initial dealer purchase transaction.
Treasury yield	Closing yield (bps) of the maturity matched treasury bond on the day prior to the initial dealer purchase transaction (with maturities in 6 months, 1, 2, 3, 5, 7, 10, and 20 years).
VIX change	Monthly change in stock market's expectation of volatility implied by S&P 500 Index options published by CBOE.